

Date: 13/12/24

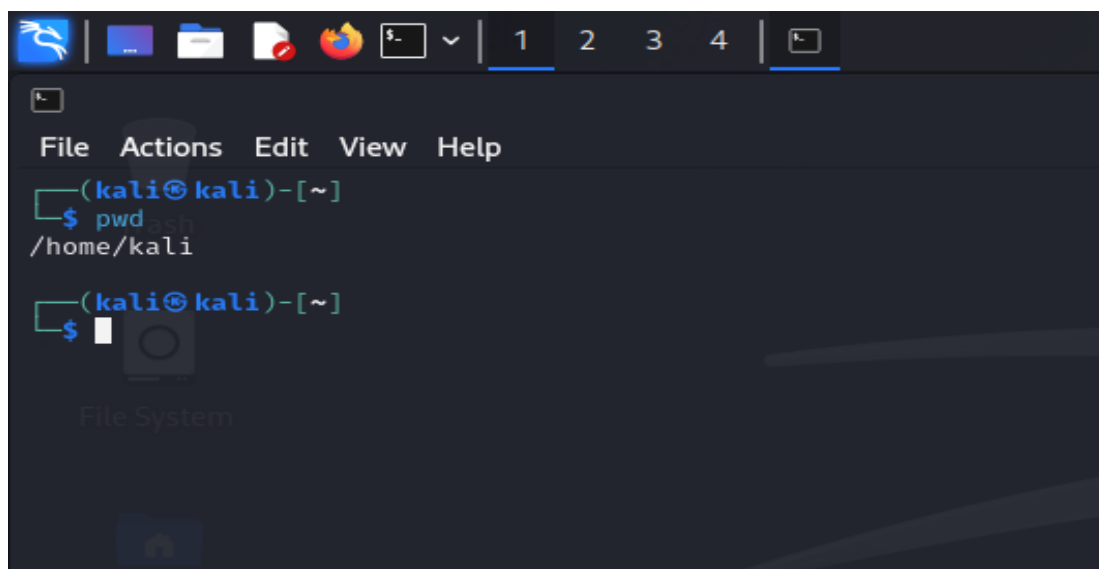
Lab Practical #01:

Perform various Linux commands and configure kali Linux with Virtual Machine.

1. pwd:

- The pwd command in Kali Linux stands for **print working directory**. It is used to display the full path of the current working directory where the user is located in the filesystem.

Implementation :



```
(kali㉿kali)-[~]  
$ pwd  
/home/kali  
  
(kali㉿kali)-[~]  
$
```

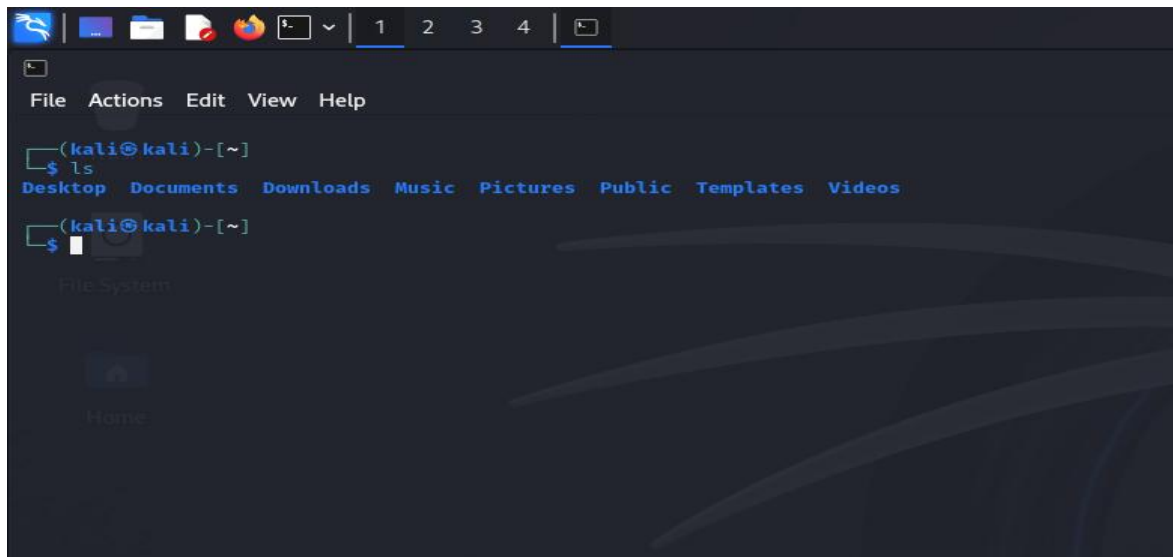
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2. ls:

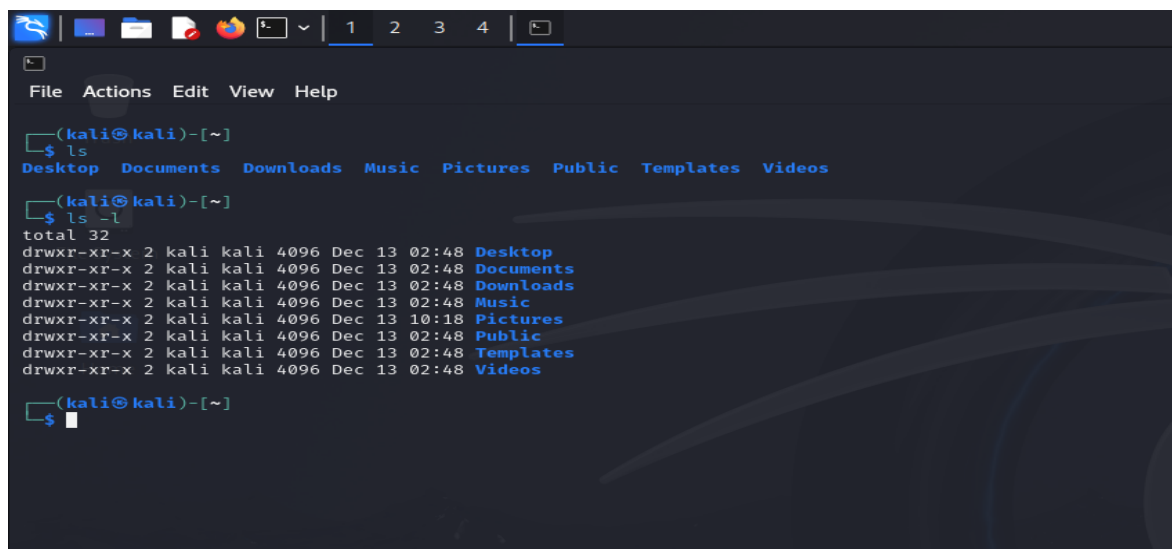
- The ls command in Kali Linux is used to **list directory contents**, including files and subdirectories. It is one of the most commonly used commands to inspect the contents of a directory.

No.	Option	Description
1	ls -l	Displays detailed information about files and directories.
2	ls -a	Shows all files, including hidden files.
3	ls -lh	Displays file sizes in human-readable format (e.g., KB, MB, GB). Used with -l.

Implementation :



```
(kali@kali)-[~]
$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
```



```
(kali@kali)-[~]
$ ls -l
total 32
drwxr-xr-x 2 kali kali 4096 Dec 13 02:48 Desktop
drwxr-xr-x 2 kali kali 4096 Dec 13 02:48 Documents
drwxr-xr-x 2 kali kali 4096 Dec 13 02:48 Downloads
drwxr-xr-x 2 kali kali 4096 Dec 13 02:48 Music
drwxr-xr-x 2 kali kali 4096 Dec 13 10:18 Pictures
drwxr-xr-x 2 kali kali 4096 Dec 13 02:48 Public
drwxr-xr-x 2 kali kali 4096 Dec 13 02:48 Templates
drwxr-xr-x 2 kali kali 4096 Dec 13 02:48 Videos
```

```
(kali@kali)-[~]
$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos

(kali@kali)-[~]
$ ls -l
total 32
drwxr-xr-x 2 kali kali 4096 Dec 13 02:48 Desktop
drwxr-xr-x 2 kali kali 4096 Dec 13 02:48 Documents
drwxr-xr-x 2 kali kali 4096 Dec 13 02:48 Downloads
drwxr-xr-x 2 kali kali 4096 Dec 13 02:48 Music
drwxr-xr-x 2 kali kali 4096 Dec 13 10:18 Pictures
drwxr-xr-x 2 kali kali 4096 Dec 13 02:48 Public
drwxr-xr-x 2 kali kali 4096 Dec 13 02:48 Templates
drwxr-xr-x 2 kali kali 4096 Dec 13 02:48 Videos

(kali@kali)-[~]
$ ls -a
.  .bashrc.original  .dmrc  .face-icon  .local  Public  .vboxclient-display-svga-x11-tty7-control.pid  .vboxclient-hostversion-tty7-control.pid  Videos  .zshrc
.  .bash_logout  .config  Downloads  .group  Music  Templates  .vboxclient-display-svga-x11-tty7-service.pid  .vboxclient-seamless-tty7-control.pid  .vboxclient-seamless-tty7-service.pid
.bashrc  Desktop  .face  .java  .profile  .vboxclient-clipboard-tty7-control.pid  .vboxclient-draganddrop-tty7-control.pid  .vboxclient-seamless-tty7-service.pid  .xsession-errors
.bashrc  Desktop  .face  .java  .profile  .vboxclient-clipboard-tty7-service.pid  .vboxclient-draganddrop-tty7-service.pid  .vboxclient-vmsvga-session-tty7-control.pid  .xsession-errors.old

(kali@kali)-[~]
$
```

```
(kali@kali)-[~]
$ ls -lh
total 32K
drwxr-xr-x 2 kali kali 4.0K Dec 13 02:48 Desktop
drwxr-xr-x 2 kali kali 4.0K Dec 13 02:48 Documents
drwxr-xr-x 2 kali kali 4.0K Dec 13 02:48 Downloads
drwxr-xr-x 2 kali kali 4.0K Dec 13 02:48 Music
drwxr-xr-x 2 kali kali 4.0K Dec 13 10:18 Pictures
drwxr-xr-x 2 kali kali 4.0K Dec 13 02:48 Public
drwxr-xr-x 2 kali kali 4.0K Dec 13 02:48 Templates
drwxr-xr-x 2 kali kali 4.0K Dec 13 02:48 Videos

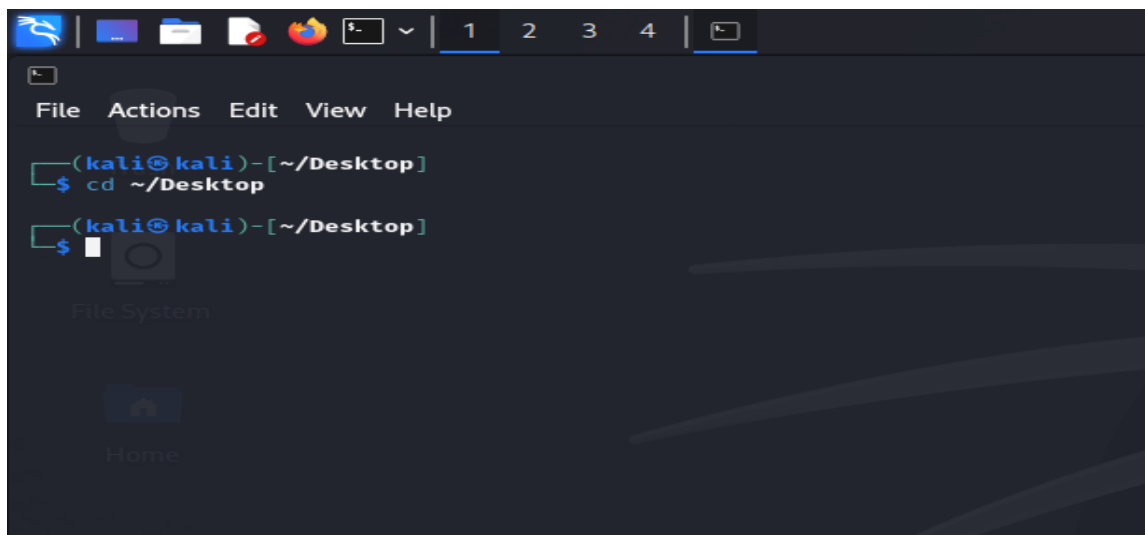
(kali@kali)-[~]
$
```

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3. cd:

- The cd command in Kali Linux is used to **change the current working directory**. It allows you to navigate through the directory structure of the filesystem.

Implementation:

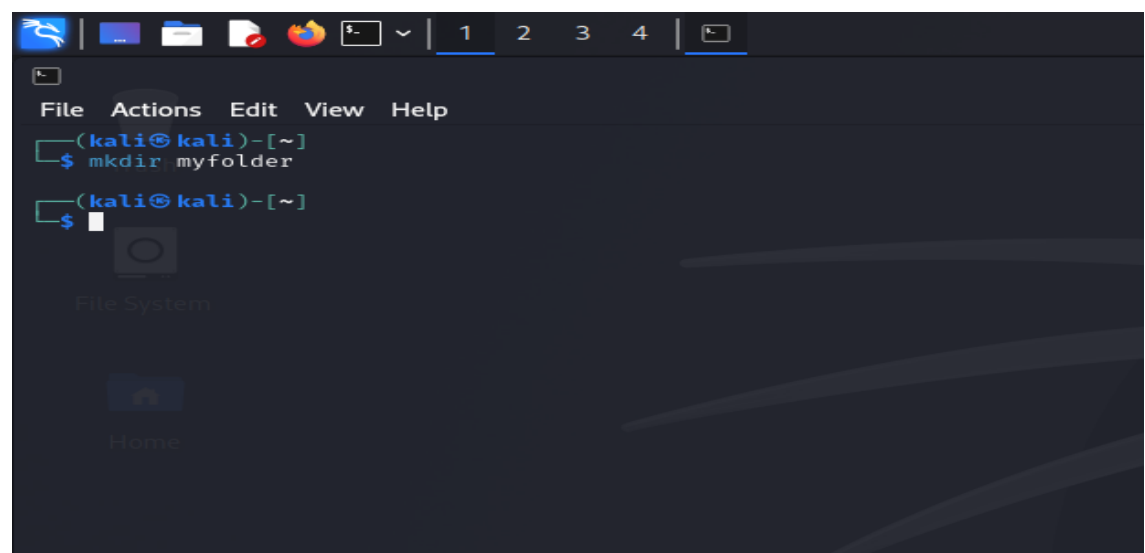


```
(kali㉿kali)-[~/Desktop]
$ cd ~/Desktop
(kali㉿kali)-[~/Desktop]
$
```

4. mkdir:

- The mkdir command in Kali Linux is used to create directories. It allows users to organize files by creating folders.

Implementation :



```
(kali㉿kali)-[~]
$ mkdir myfolder
(kali㉿kali)-[~]
$
```

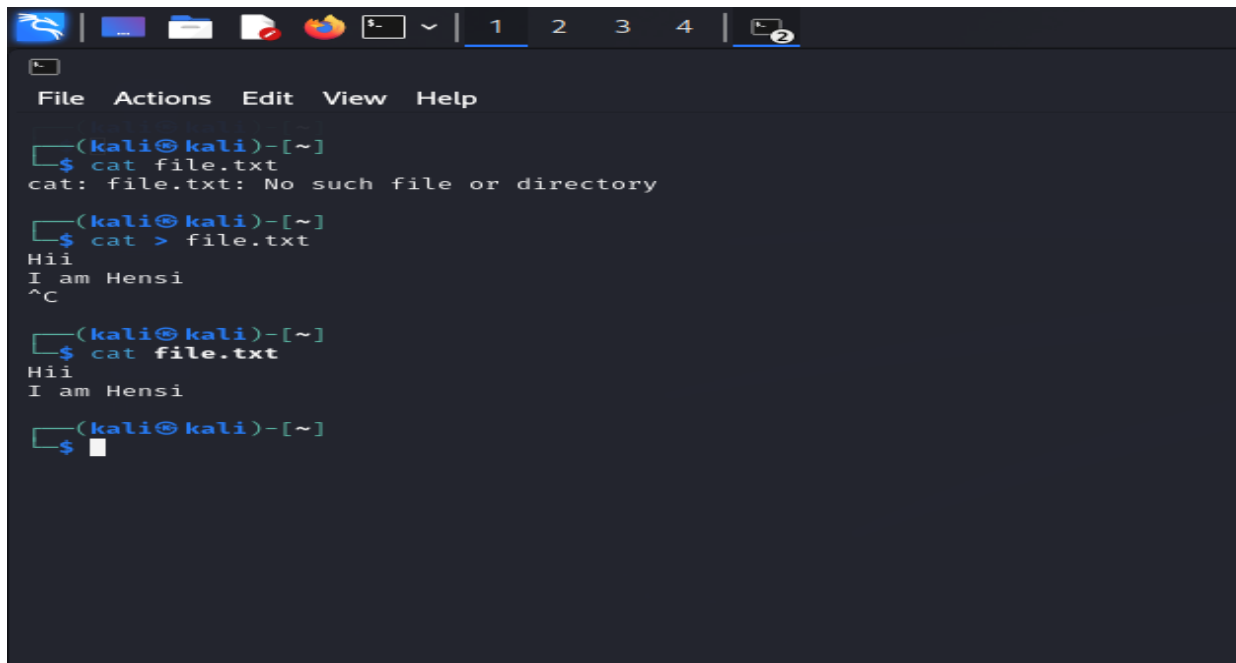
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5. Cat:

- The cat command in Kali Linux (and other Linux distributions) is used to **display the contents of files, combine multiple files, and create or append to files**. It is one of the most commonly used commands for viewing file content.

No.	Option	Description
1	cat -n file.txt	Numbers all lines of output.
2	cat -E file.txt	Displays a \$ at the end of each line to indicate line endings.
3	cat -b file.txt	Number non-blank output lines.

Implementation :



```
(kali㉿kali)-[~]
$ cat file.txt
cat: file.txt: No such file or directory

(kali㉿kali)-[~]
$ cat > file.txt
Hii
I am Hensi
^C

(kali㉿kali)-[~]
$ cat file.txt
Hii
I am Hensi

(kali㉿kali)-[~]
$
```



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```
File Actions Edit View Help
(kali㉿kali)-[~]
$ cat -n file.txt
1 Hii
2 I am Hensi
cat: 2: I am Hensi: such file or directory

(kali㉿kali)-[~]
$ cat file.txt
Hii
I am Hensi
^C

(kali㉿kali)-[~]
$ cat file.txt
Hii
I am Hensi
```

```
File Actions Edit View Help
(kali㉿kali)-[~]
$ cat -E file.txt
Hii$
I am Hensi$

(kali㉿kali)-[~]
$ cat file.txt
Hii
2 I am Hensi

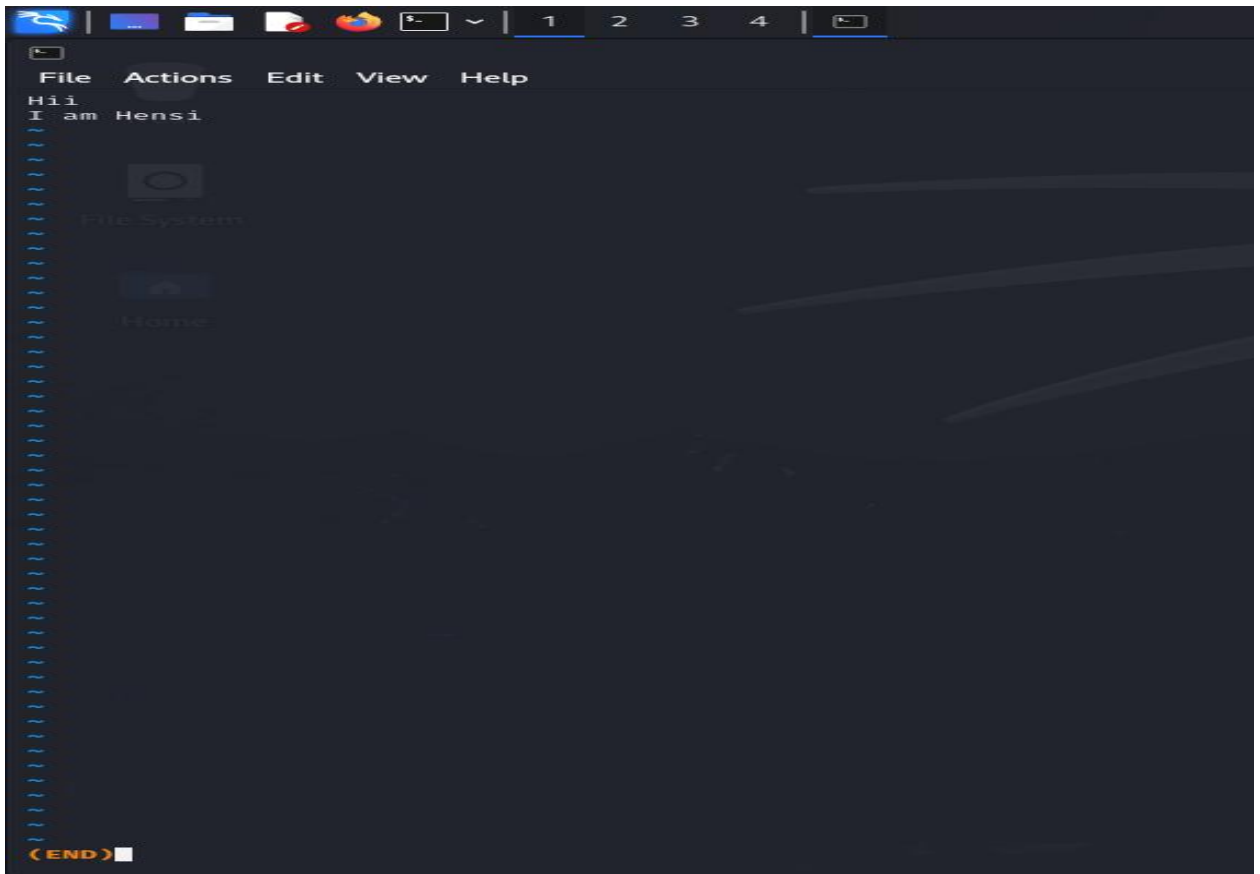
(kali㉿kali)-[~]
$
```

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6. less:

- The less command in Kali Linux is used to view the content of files one screen at a time. It is especially useful for large files, as it does not load the entire file into memory, unlike the cat command.

Implementation :

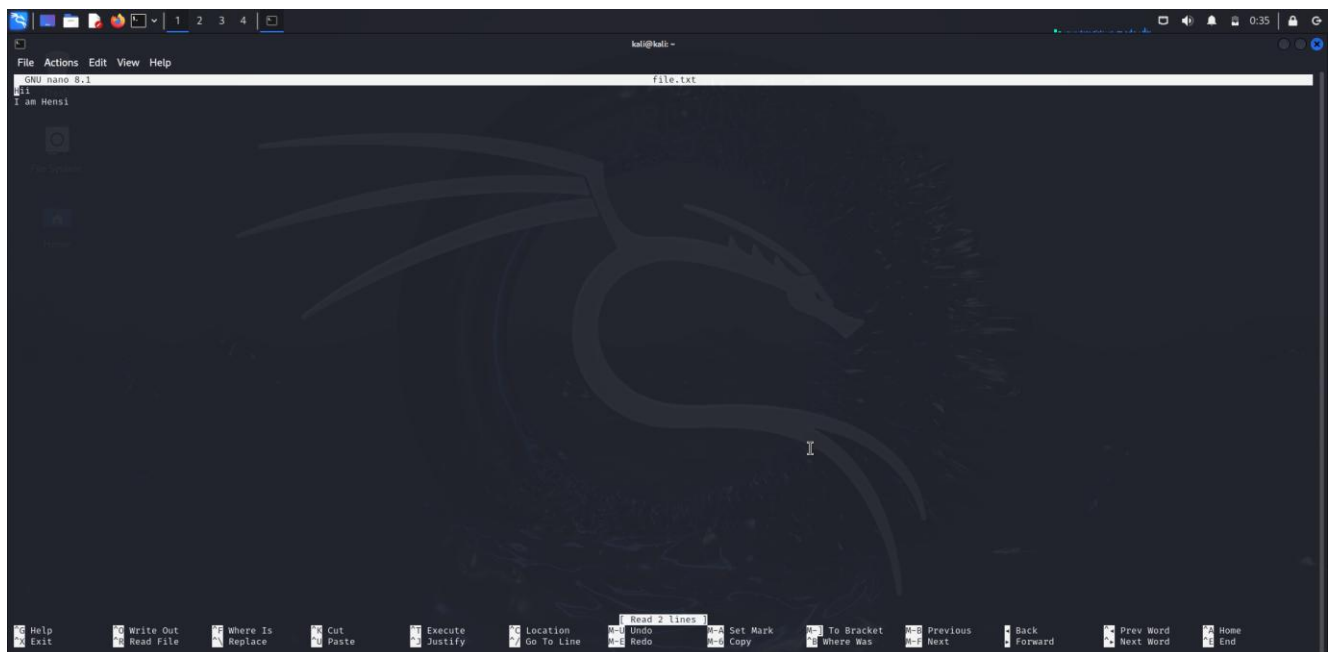


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7. nano:

- The nano command in Kali Linux is a simple and user-friendly text editor used to create and edit files directly from the terminal. It is widely used due to its ease of use and straightforward interface.

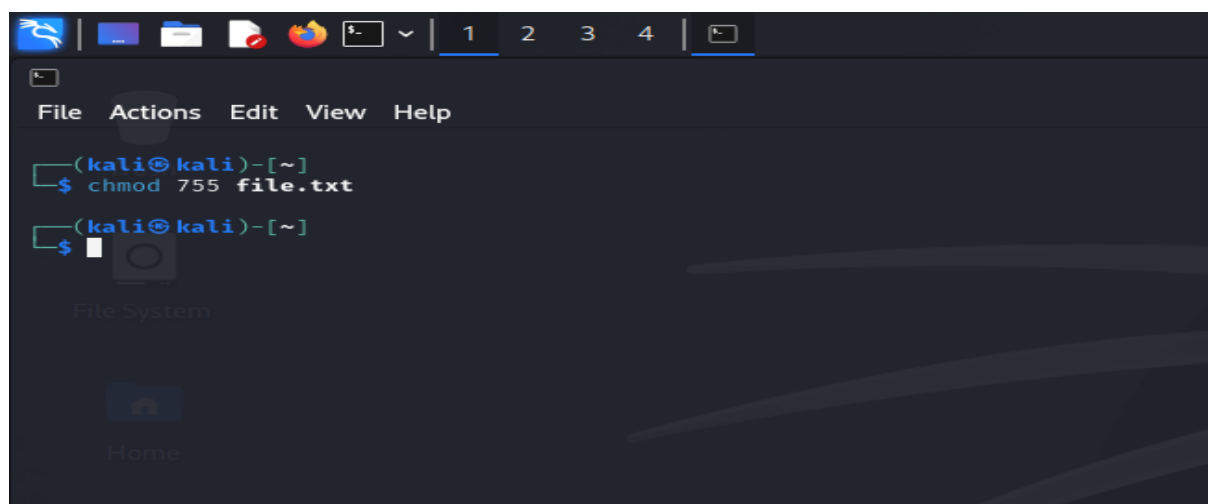
Implementation :



8. chmod:

- The chmod command in Kali Linux is used **to change the permissions** of files and directories. File permissions determine who can read, write, or execute a file or directory.

Implementation :



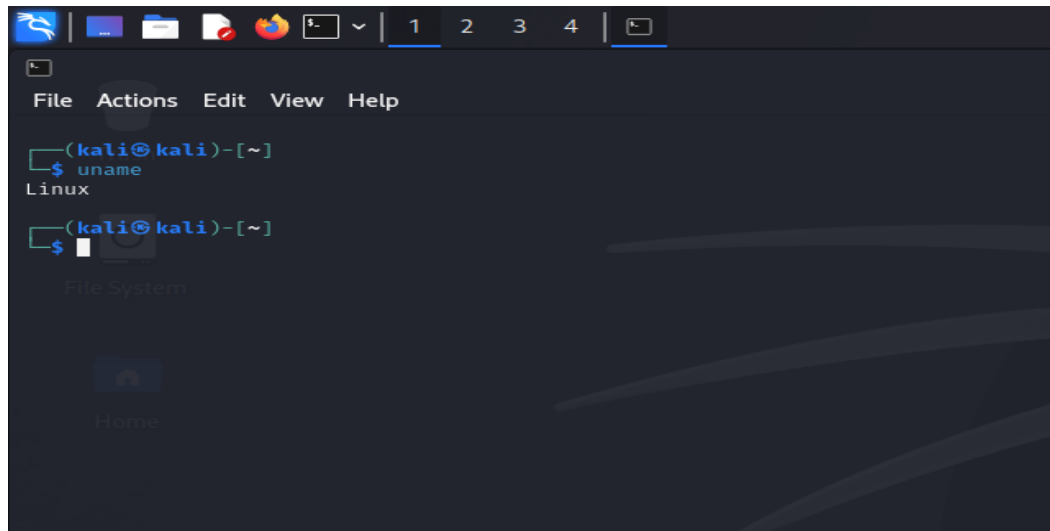
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9. uname:

- The uname command in Kali Linux, is used to display **system information** such as the kernel name, version, architecture, and more. It provides a quick way to check details about your operating system and kernel.

No.	Option	Description
1	uname -a	Display all available information.
2	uname -n	Show the network hostname.
3	uname -v	Show the kernel version.
4	uname -p	Show the processor type (may display "unknown").

Implementation :

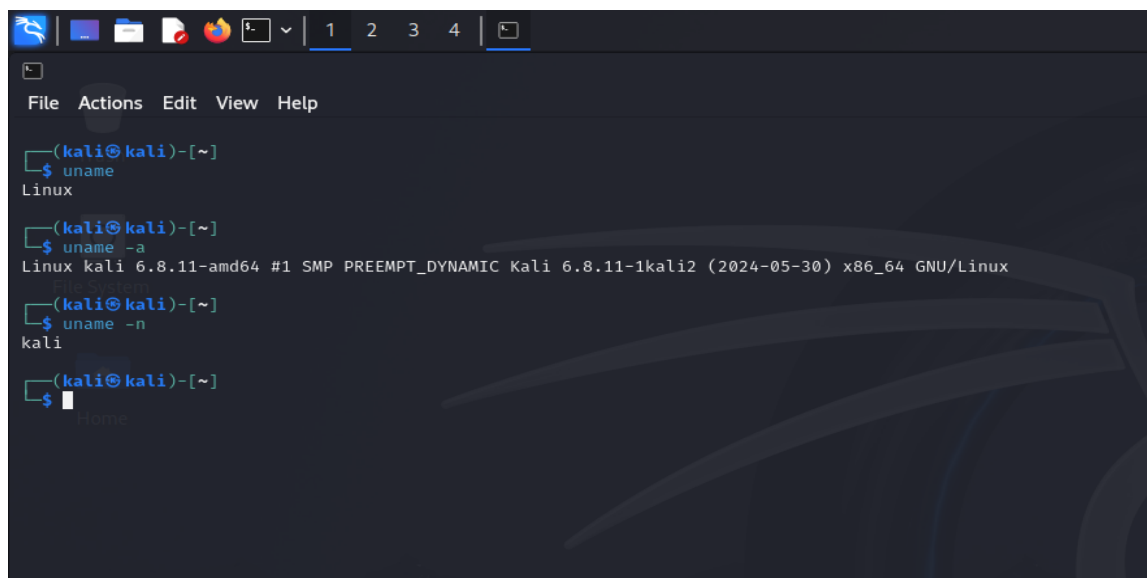


```
(kali㉿kali)-[~]  
$ uname  
Linux  
(kali㉿kali)-[~]  
$
```

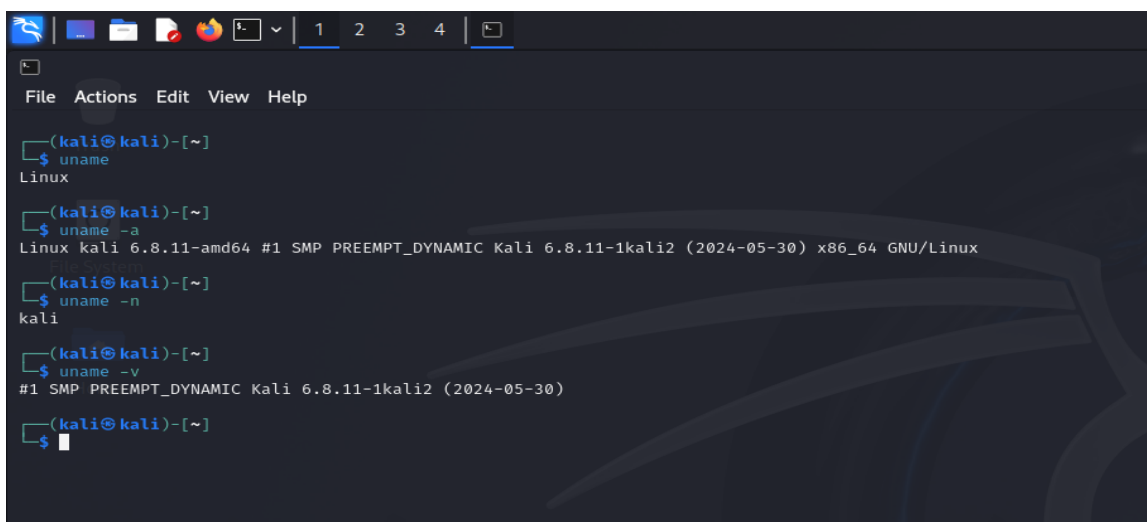
Date: 13/12/24



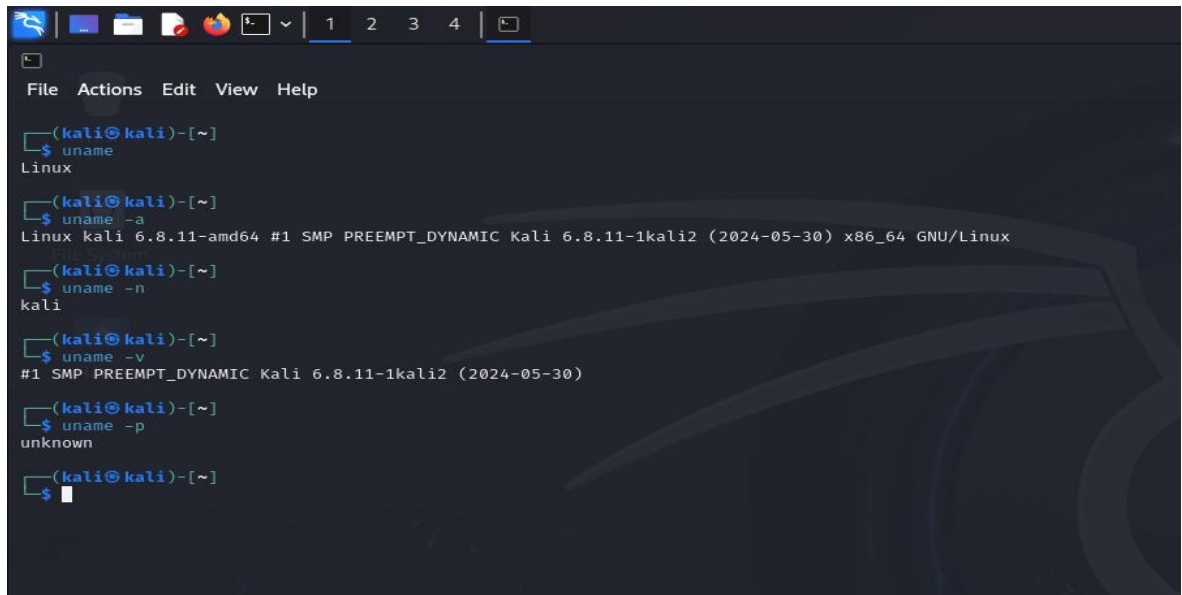
```
(kali㉿kali)-[~]  
$ uname  
Linux  
  
(kali㉿kali)-[~]  
$ uname -a  
Linux kali 6.8.11-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.8.11-1kali2 (2024-05-30) x86_64 GNU/Linux  
File System  
(kali㉿kali)-[~]  
$
```



```
(kali㉿kali)-[~]  
$ uname  
Linux  
  
(kali㉿kali)-[~]  
$ uname -a  
Linux kali 6.8.11-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.8.11-1kali2 (2024-05-30) x86_64 GNU/Linux  
File System  
(kali㉿kali)-[~]  
$ uname -n  
kali  
  
(kali㉿kali)-[~]  
$
```



```
(kali㉿kali)-[~]  
$ uname  
Linux  
  
(kali㉿kali)-[~]  
$ uname -a  
Linux kali 6.8.11-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.8.11-1kali2 (2024-05-30) x86_64 GNU/Linux  
File System  
(kali㉿kali)-[~]  
$ uname -n  
kali  
  
(kali㉿kali)-[~]  
$ uname -v  
#1 SMP PREEMPT_DYNAMIC Kali 6.8.11-1kali2 (2024-05-30)  
  
(kali㉿kali)-[~]  
$
```

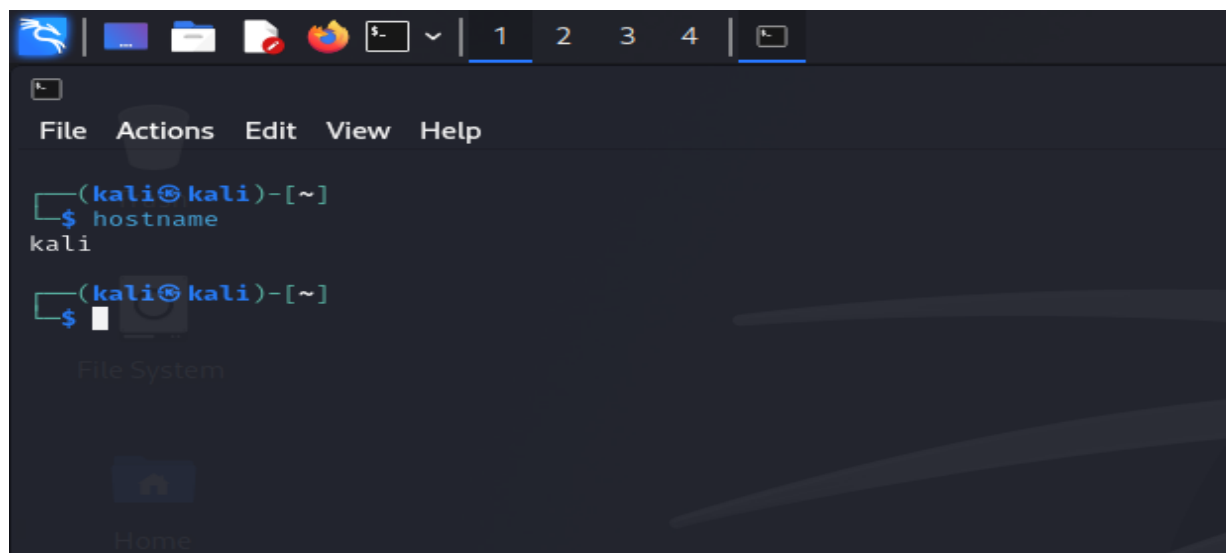


```
(kali@kali)~  
$ uname  
Linux  
(kali@kali)~  
$ uname -a  
Linux kali 6.8.11-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.8.11-1kali2 (2024-05-30) x86_64 GNU/Linux  
(kali@kali)~  
$ uname -n  
kali  
(kali@kali)~  
$ uname -v  
#1 SMP PREEMPT_DYNAMIC Kali 6.8.11-1kali2 (2024-05-30)  
(kali@kali)~  
$ uname -p  
unknown  
(kali@kali)~  
$
```

10. hostname:

- The hostname command in Linux is used to display or set the **hostname** of the system. A hostname is a unique name that identifies a computer on a network.

Implementation :



```
(kali@kali)~  
$ hostname  
kali  
(kali@kali)~  
$
```

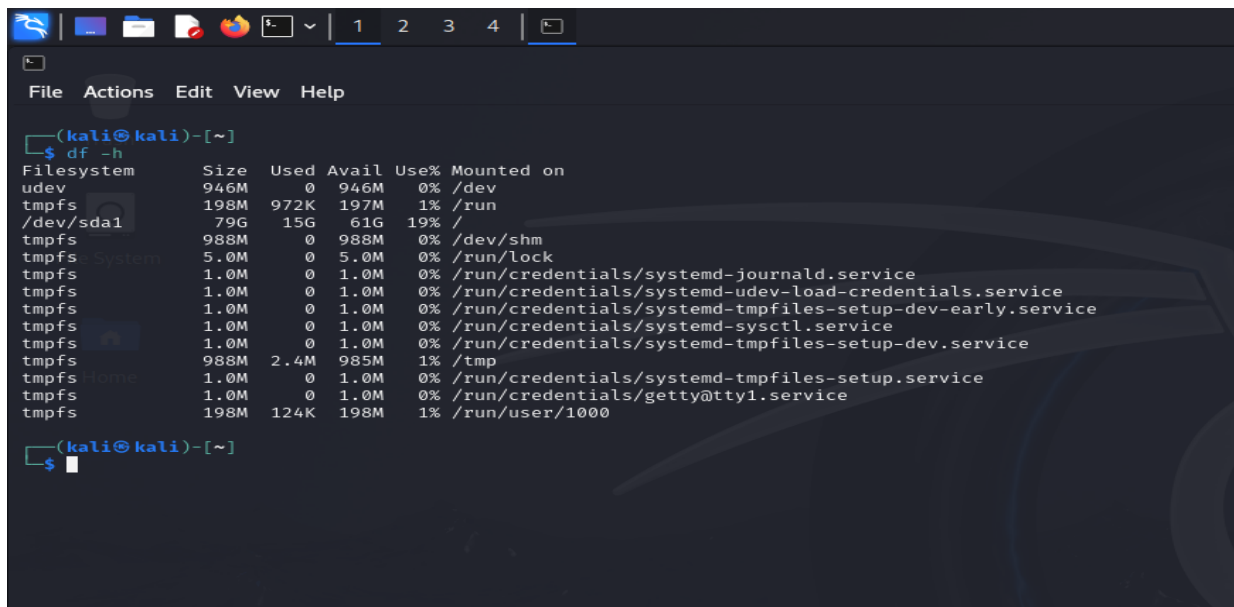
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11. df:

- The df command in Linux is used to display **disk space usage** of file systems. It shows the available and used space for each mounted file system on the system.

No.	Option	Description
1	df -h	Display Disk Usage in Human-Readable Format.
2	df -i	Show inode usage instead of block usage.
3	df -T	Show the file system type.

Implementation :



```
(kali@kali)~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            946M   0    946M   0% /dev
tmpfs           198M  972K   197M   1% /run
/dev/sda1       79G   15G   61G   19% /
tmpfs           988M   0    988M   0% /dev/shm
tmpfs           5.0M   0    5.0M   0% /run/lock
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-journald.service
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-udev-load-credentials.service
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-tmpfiles-setup-dev-early.service
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-sysctl.service
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-tmpfiles-setup-dev.service
tmpfs           988M  2.4M   985M   1% /tmp
tmpfs           1.0M   0    1.0M   0% /run/credentials/systemd-tmpfiles-setup.service
tmpfs           1.0M   0    1.0M   0% /run/credentials/getty@tty1.service
tmpfs           198M  124K   198M   1% /run/user/1000
```

```

(kali@kali)-[~]
$ df -i
Filesystem            Inodes    IUsed    IFree   IUse% Mounted on
udev                  242042      391    241651      1% /dev
tmpfs                 252745      640    252105      1% /run
/dev/sda1             5251072  445216  4805856      9% /
tmpfs                 252745      1    252744      1% /dev/shm
tmpfs                 252745      2    252743      1% /run/lock
tmpfs                 1024      1    1023      1% /run/credentials/systemd-journald.service
tmpfs                 1024      1    1023      1% /run/credentials/systemd-udev-load-credentials.service
tmpfs                 1024      1    1023      1% /run/credentials/systemd-tmpfiles-setup-dev-early.service
tmpfs                 1024      1    1023      1% /run/credentials/systemd-sysctl.service
tmpfs                 1024      1    1023      1% /run/credentials/systemd-tmpfiles-setup-dev.service
tmpfs                1048576     36  1048540      1% /tmp
tmpfs                 1024      1    1023      1% /run/credentials/systemd-tmpfiles-setup.service
tmpfs                 1024      1    1023      1% /run/credentials/getty@tty1.service
tmpfs                 50549     118    50431      1% /run/user/1000
  
```

```

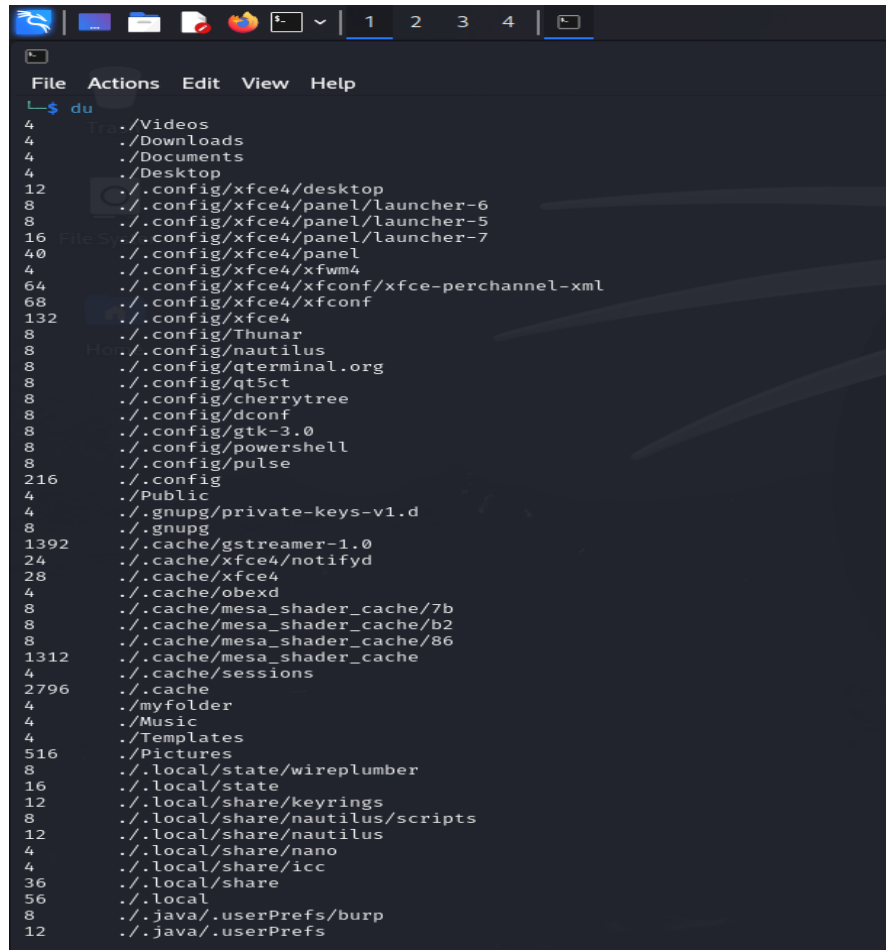
(kali@kali)-[~]
$ df -T
Filesystem            Type      1K-blocks    Used Available Use% Mounted on
udev                 devtmpfs      968168      0    968168   0% /dev
tmpfs                tmpfs        202196      972    201224   1% /run
/dev/sda1            ext4       82083148  14751692  63115908  19% /
tmpfs                tmpfs       1010980      0    1010980   0% /dev/shm
tmpfs                 tmpfs        5120      0     5120   0% /run/lock
tmpfs                 tmpfs        1024      0     1024   0% /run/credentials/systemd-journald.service
tmpfs                 tmpfs        1024      0     1024   0% /run/credentials/systemd-udev-load-credentials.service
tmpfs                 tmpfs        1024      0     1024   0% /run/credentials/systemd-tmpfiles-setup-dev-early.service
tmpfs                 tmpfs        1024      0     1024   0% /run/credentials/systemd-sysctl.service
tmpfs                 tmpfs        1024      0     1024   0% /run/credentials/systemd-tmpfiles-setup-dev.service
tmpfs                 tmpfs       1010980    3024    1007956   1% /tmp
tmpfs                 tmpfs        1024      0     1024   0% /run/credentials/systemd-tmpfiles-setup.service
tmpfs                 tmpfs        1024      0     1024   0% /run/credentials/getty@tty1.service
tmpfs                 tmpfs       202196     124    202072   1% /run/user/1000
  
```

12. du:

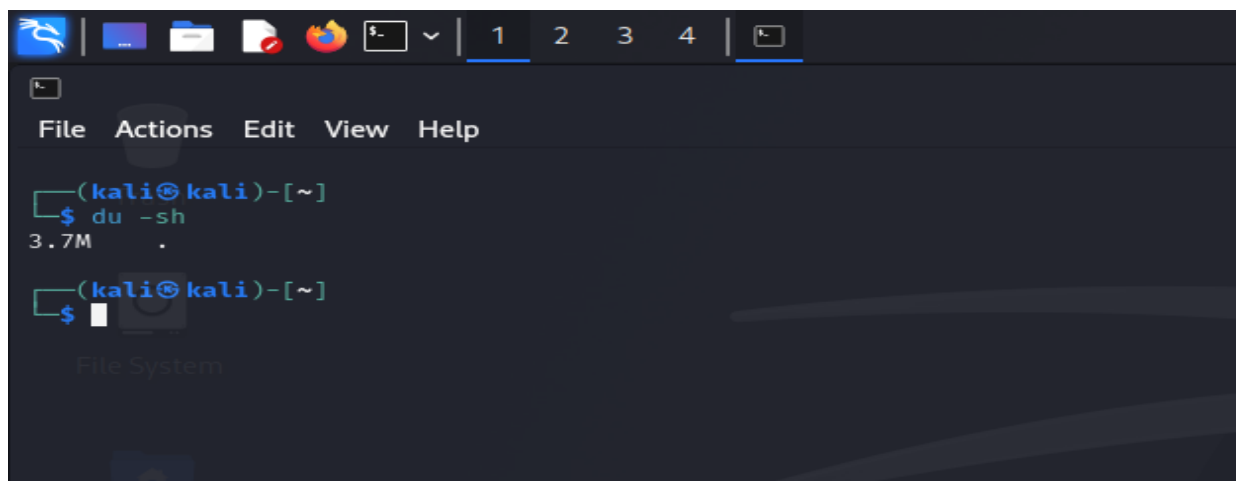
- The du command in Linux is used to estimate and display the **disk usage** of files and directories. It provides detailed information about how much space files or directories consume on the filesystem.

No.	Option	Description
1	du -sh	Display Disk Usage in Human-Readable Format.
2	du -s	Show only the total size for each argument.
3	du -h --max-depth=2	Show Disk Usage with Total Size for Each Level of Subdirectories.
4	du -ah --exclude=*.log	Display Disk Usage for Each Directory and File.

Implementation :

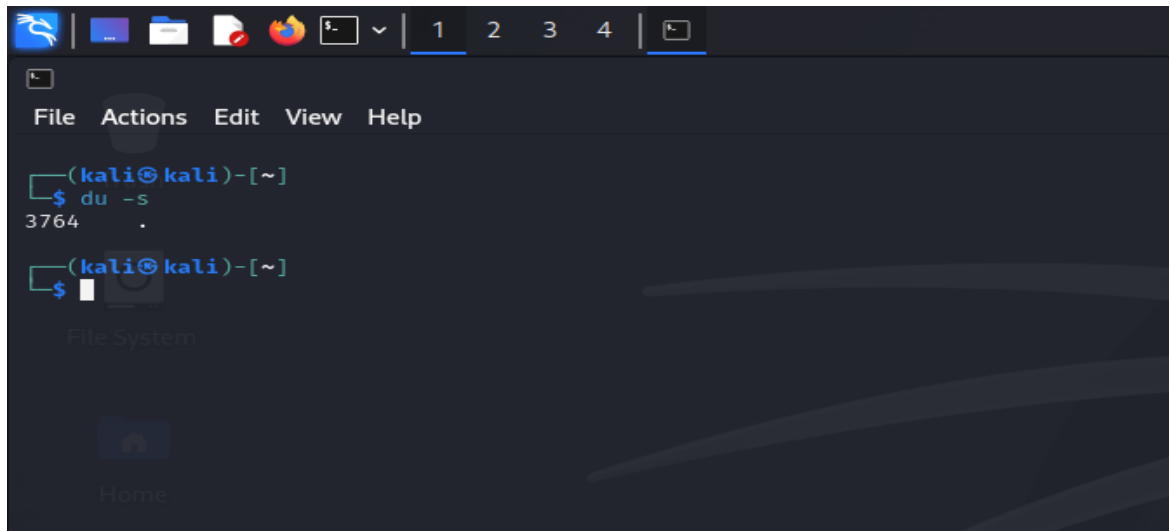


```
File Actions Edit View Help
└─$ du
4 ./.Videos
4 ./.Downloads
4 ./.Documents
4 ./.Desktop
12 ./.config/xfce4/desktop
8 ./.config/xfce4/panel/launcher-6
8 ./.config/xfce4/panel/launcher-5
16 ./.config/xfce4/panel/launcher-7
40 ./.config/xfce4/panel
4 ./.config/xfce4/xfwm4
64 ./.config/xfce4/xfconf/xfce-perchannel-xml
68 ./.config/xfce4/xfconf
132 ./.config/xfce4
8 ./.config/Thunar
8 ./.config/nautilus
8 ./.config/qterminal.org
8 ./.config/qt5ct
8 ./.config/cherrytree
8 ./.config/dconf
8 ./.config/gtk-3.0
8 ./.config/powershell
8 ./.config/pulse
216 ./.config
4 ./.Public
4 ./.gnupg/private-keys-v1.d
8 ./.gnupg
1392 ./.cache/gstreamer-1.0
24 ./.cache/xfce4/notifyd
28 ./.cache/xfce4
4 ./.cache/obexd
8 ./.cache/mesa_shader_cache/7b
8 ./.cache/mesa_shader_cache/b2
8 ./.cache/mesa_shader_cache/86
1312 ./.cache/mesa_shader_cache
4 ./.cache/sessions
2796 ./.cache
4 ./.myfolder
4 ./.Music
4 ./.Templates
516 ./.Pictures
8 ./.local/state/wireplumber
16 ./.local/state
12 ./.local/share/keyrings
8 ./.local/share/nautilus/scripts
12 ./.local/share/nautilus
4 ./.local/share/nano
4 ./.local/share/icc
36 ./.local/share
56 ./.local
8 ./.java/.userPrefs/burp
12 ./.java/.userPrefs
```

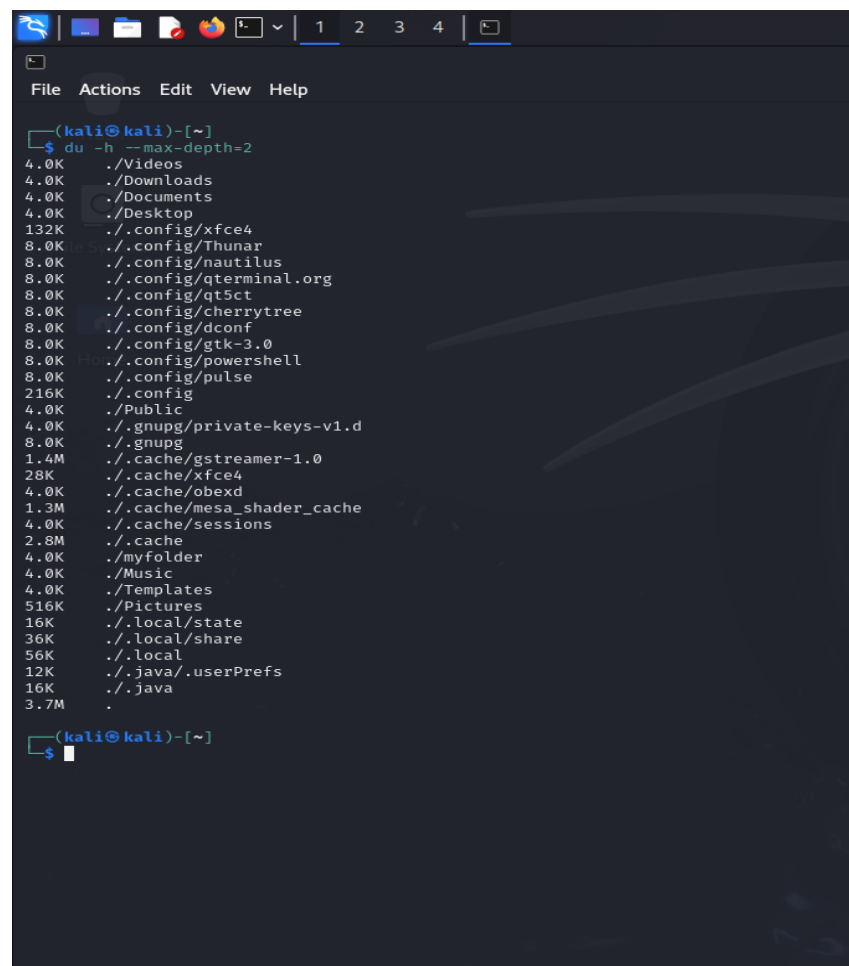


```
File Actions Edit View Help
└─(kali㉿kali)-[~]
└─$ du -sh
3.7M .
└─(kali㉿kali)-[~]
└─$
```

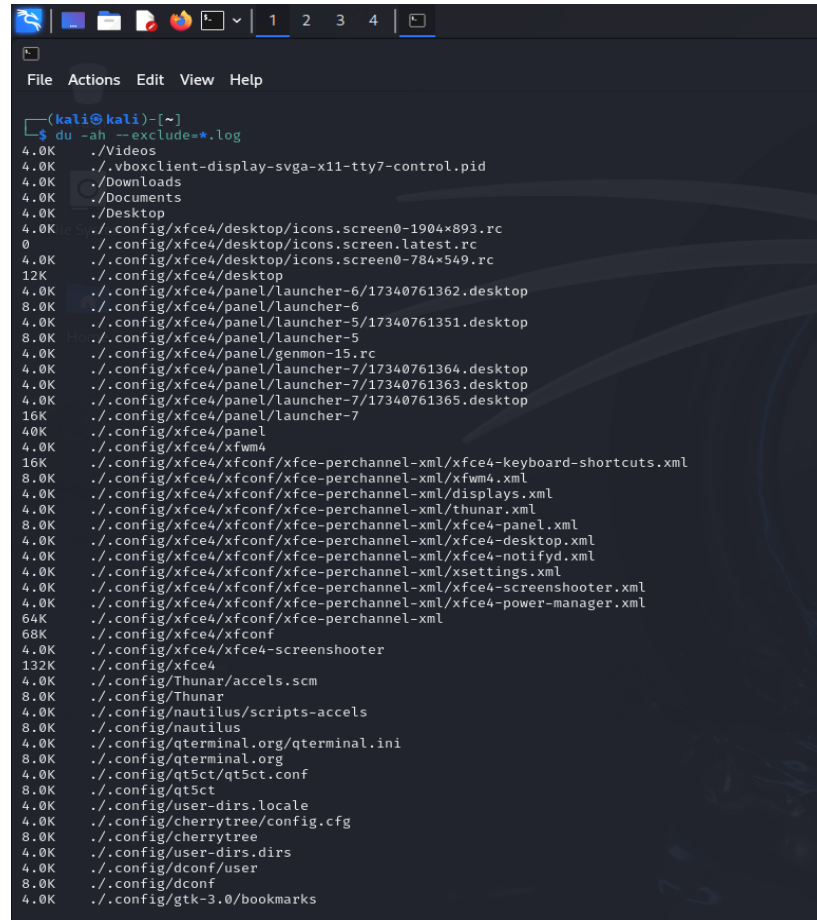
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```
(kali㉿kali)-[~]  
$ du -s  
3764 .  
  
(kali㉿kali)-[~]  
$
```



```
(kali㉿kali)-[~]  
$ du -h --max-depth=2  
4.0K    ./Videos  
4.0K    ./Downloads  
4.0K    ./Documents  
4.0K    ./Desktop  
132K    ./config/xfce4  
8.0K    ./config/Thunar  
8.0K    ./config/nautilus  
8.0K    ./config/qterminal.org  
8.0K    ./config/qt5ct  
8.0K    ./config/cherrytree  
8.0K    ./config/dconf  
8.0K    ./config/gtk-3.0  
8.0K    ./config/powershell  
8.0K    ./config/pulse  
216K    ./config  
4.0K    ./Public  
4.0K    ./gnupg/private-keys-v1.d  
8.0K    ./gnupg  
1.4M    ./cache/gstreamer-1.0  
28K    ./cache/xfce4  
4.0K    ./cache/obexd  
1.3M    ./cache/mesa_shader_cache  
4.0K    ./cache/sessions  
2.8M    ./cache  
4.0K    ./myfolder  
4.0K    ./Music  
4.0K    ./Templates  
516K    ./Pictures  
16K    ./local/state  
36K    ./local/share  
56K    ./local  
12K    ./java/.userPrefs  
16K    ./java  
3.7M    .  
  
(kali㉿kali)-[~]  
$
```



```

(kali@kali)-[~]
$ du -ah --exclude=*.log
4.0K  ./Videos
4.0K  ./vboxclient-display-svga-x11-tty7-control.pid
4.0K  ./Downloads
4.0K  ./Documents
4.0K  ./Desktop
4.0K  ./config/xfce4/desktop/icons.screen0-1904x893.rc
0     ./config/xfce4/desktop/icons.screen.latest.rc
4.0K  ./config/xfce4/desktop/icons.screen0-784x549.rc
12K  ./config/xfce4/desktop
4.0K  ./config/xfce4/panel/launcher-6/17340761362.desktop
8.0K  ./config/xfce4/panel/launcher-6
4.0K  ./config/xfce4/panel/launcher-5/17340761351.desktop
8.0K  ./config/xfce4/panel/launcher-5
4.0K  ./config/xfce4/panel/genmon-15.rc
4.0K  ./config/xfce4/panel/launcher-7/17340761364.desktop
4.0K  ./config/xfce4/panel/launcher-7/17340761363.desktop
4.0K  ./config/xfce4/panel/launcher-7/17340761365.desktop
16K  ./config/xfce4/panel/launcher-7
4.0K  ./config/xfce4/panel
4.0K  ./config/xfce4/xfwm4
16K  ./config/xfce4/xfconf/xfce-perchannel-xml/xfce4-keyboard-shortcuts.xml
8.0K  ./config/xfce4/xfconf/xfce-perchannel-xml/xfwm4.xml
4.0K  ./config/xfce4/xfconf/xfce-perchannel-xml/displays.xml
4.0K  ./config/xfce4/xfconf/xfce-perchannel-xml/thunar.xml
8.0K  ./config/xfce4/xfconf/xfce-perchannel-xml/xfce4-panel.xml
4.0K  ./config/xfce4/xfconf/xfce-perchannel-xml/xfce4-desktop.xml
4.0K  ./config/xfce4/xfconf/xfce-perchannel-xml/xfce4-notifyd.xml
4.0K  ./config/xfce4/xfconf/xfce-perchannel-xml/xsettings.xml
4.0K  ./config/xfce4/xfconf/xfce-perchannel-xml/xfce4-screenshooter.xml
4.0K  ./config/xfce4/xfconf/xfce-perchannel-xml/xfce4-power-manager.xml
64K  ./config/xfce4/xfconf/xfce-perchannel-xml
68K  ./config/xfce4/xfconf
4.0K  ./config/xfce4/xfce4-screenshooter
132K ./config/xfce4
4.0K  ./config/Thunar/accels.scm
8.0K  ./config/Thunar
4.0K  ./config/nautilus/scripts-accels
8.0K  ./config/nautilus
4.0K  ./config/qterminal.org/qterminal.ini
8.0K  ./config/qterminal.org
4.0K  ./config/qt5ct/qt5ct.conf
8.0K  ./config/qt5ct
4.0K  ./config/user-dirs.locale
4.0K  ./config/cherrytree/config.cfg
8.0K  ./config/cherrytree
4.0K  ./config/user-dirs.dirs
4.0K  ./config/dconf/user
8.0K  ./config/dconf
4.0K  ./config/gtk-3.0/bookmarks
  
```

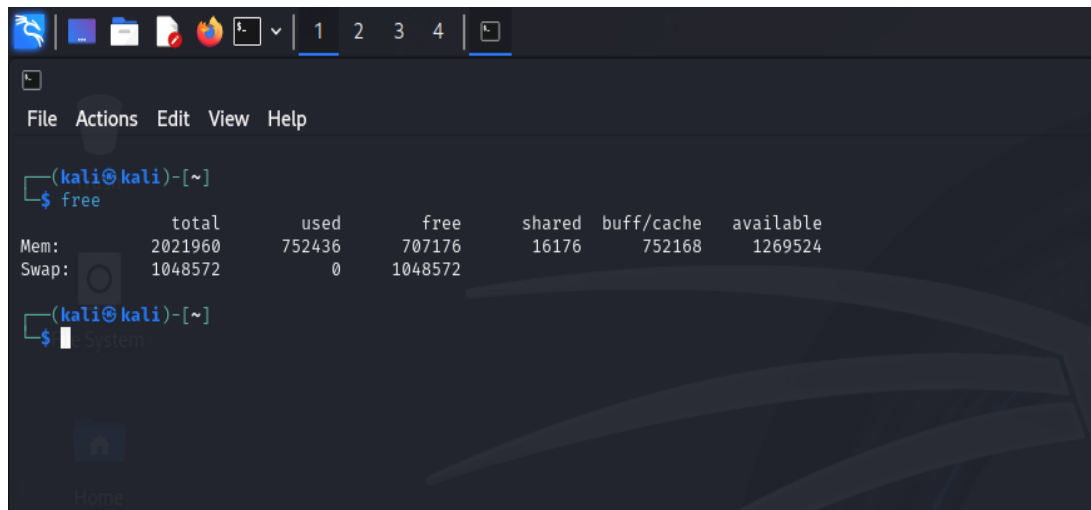
13. free:

- The free command in Kali Linux is used to display the **amount of free and used memory** in the system, including both **physical memory (RAM)** and **swap space**. It provides a quick overview of memory usage, which is crucial for monitoring system performance.

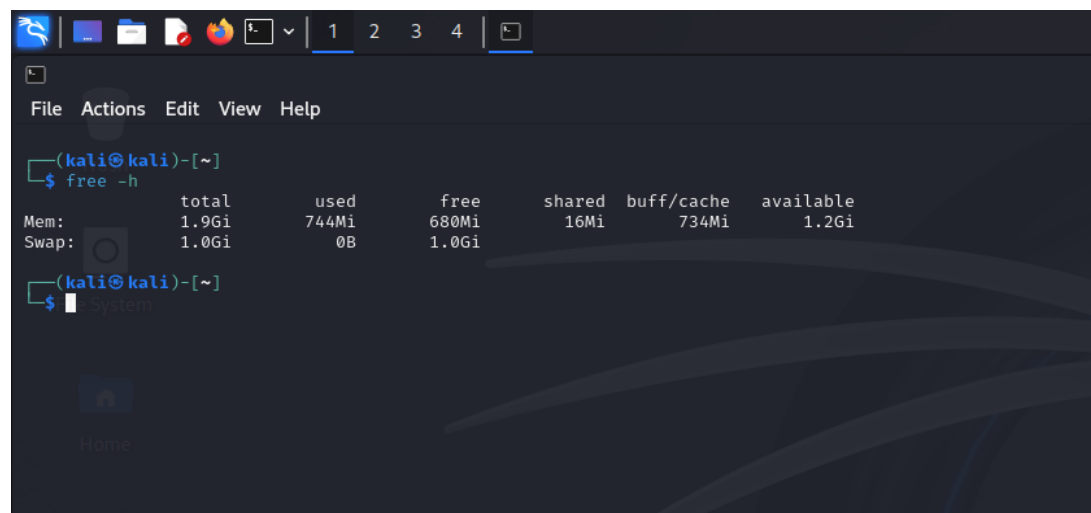
No.	Option	Description
1	free -h	Display Memory Usage in Human-Readable Format.
2	free -m	Display Memory Usage in Megabytes.
3	free -t	Display Detailed Memory Information.

Date: 13/12/24

Implementation :



```
(kali@kali)-[~]
$ free
              total        used        free      shared  buff/cache   available
Mem:           2021960       752436       707176        16176        752168       1269524
Swap:           1048572           0        1048572
```



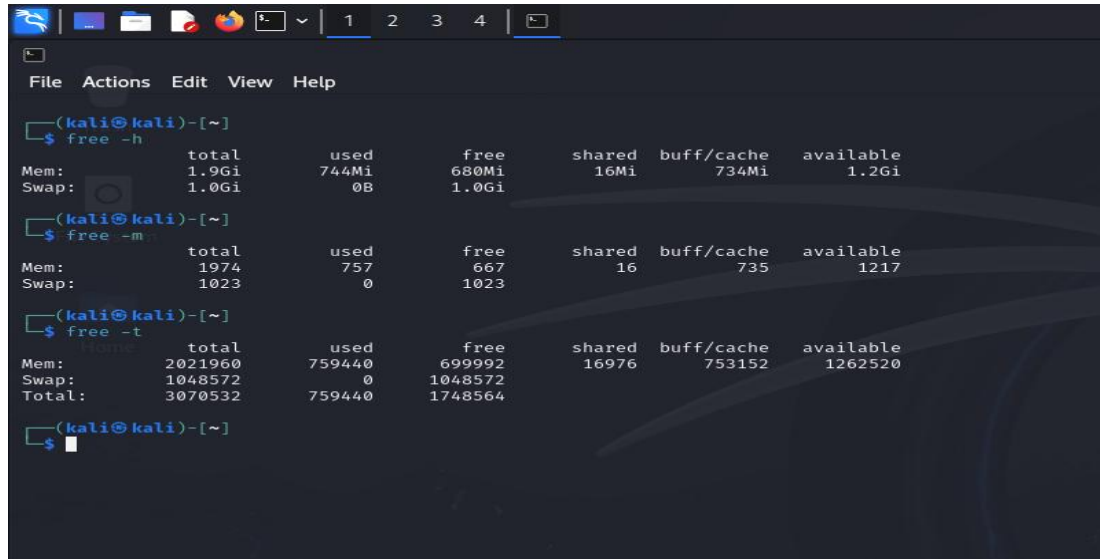
```
(kali@kali)-[~]
$ free -h
              total        used        free      shared  buff/cache   available
Mem:           1.9Gi       744Mi       680Mi        16Mi        734Mi       1.2Gi
Swap:           1.0Gi           0B        1.0Gi
```



```
(kali@kali)-[~]
$ free -h
              total        used        free      shared  buff/cache   available
Mem:           1.9Gi       744Mi       680Mi        16Mi        734Mi       1.2Gi
Swap:           1.0Gi           0B        1.0Gi

(kali@kali)-[~]
$ free -m
              total        used        free      shared  buff/cache   available
Mem:           1974         757         667         16         735        1217
Swap:           1023           0        1023
```

Date: 13/12/24



```
(kali@kali)-[~]
$ free -h
              total        used        free      shared  buff/cache   available
Mem:           1.9Gi       744Mi       680Mi       16Mi       734Mi       1.2Gi
Swap:           1.0Gi          0B       1.0Gi

(kali@kali)-[~]
$ free -m
              total        used        free      shared  buff/cache   available
Mem:           1974          757          667          16          735          1217
Swap:           1023           0          1023

(kali@kali)-[~]
$ free -t
              total        used        free      shared  buff/cache   available
Mem:           2021960       759440       699992       16976       753152       1262520
Swap:           1048572           0       1048572
Total:         3070532       759440       1748564

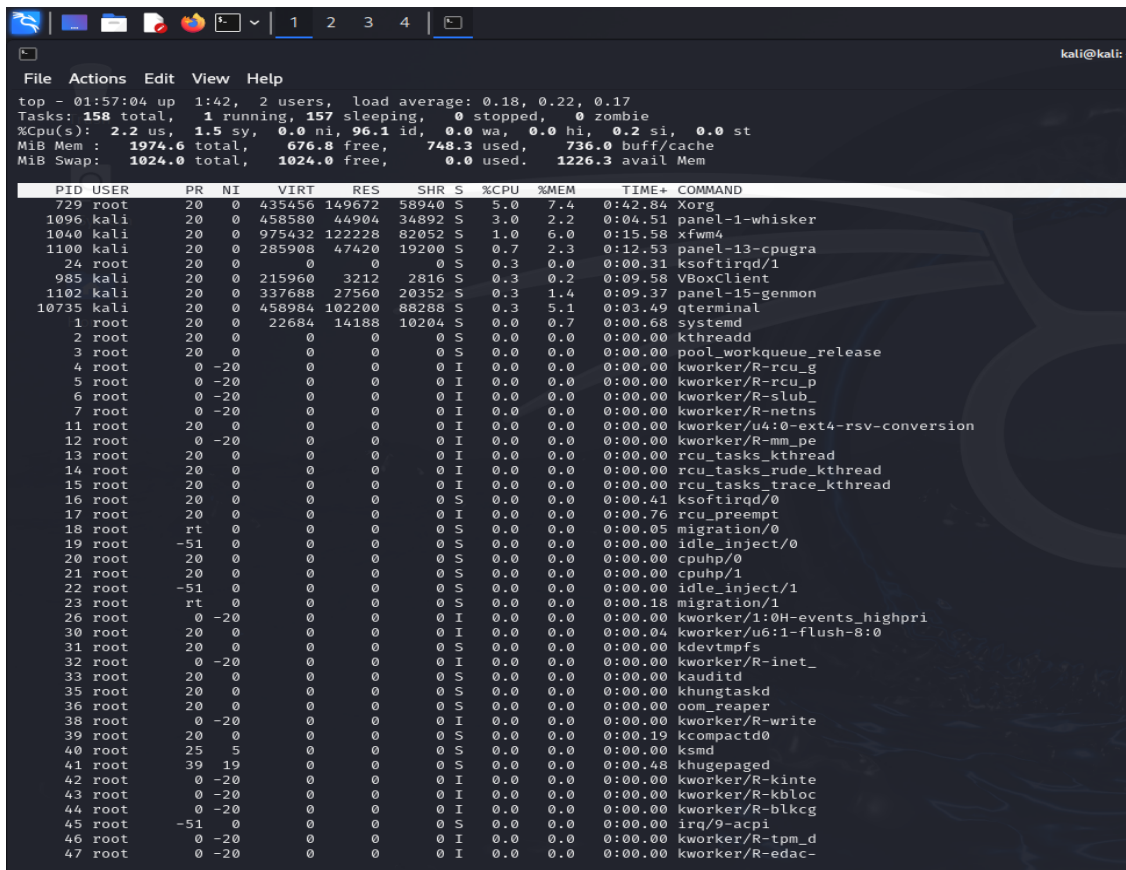
(kali@kali)-[~]
$
```

14. top:

- The top command in Kali Linux provides a real-time, dynamic view of the system's **CPU**, **memory**, and **processes** usage. It is a powerful tool to monitor system performance and resource consumption, such as CPU usage, memory usage, and the status of running processes.

No.	Option	Description
1	top -h	Display help or more information about top.

Implementation :

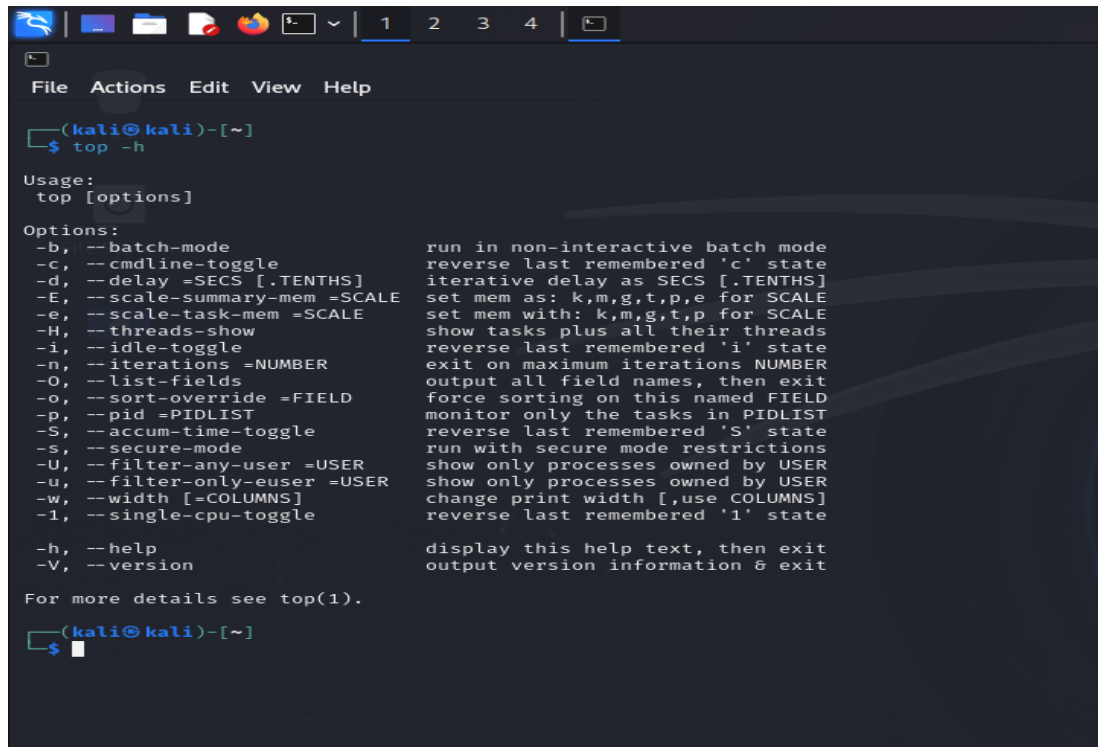


```

top - 01:57:04 up 1:42, 2 users, load average: 0.18, 0.22, 0.17
Tasks: 158 total, 1 running, 157 sleeping, 0 stopped, 0 zombie
%Cpu(s): 2.2 us, 1.5 sy, 0.0 ni, 96.1 id, 0.0 wa, 0.0 hi, 0.2 si, 0.0 st
MiB Mem : 1974.6 total, 676.8 free, 748.3 used, 736.0 buff/cache
MiB Swap: 1024.0 total, 1024.0 free, 0.0 used, 1226.3 avail Mem

  PID USER      PR  NI   VIRT  RES  SHR S %CPU  %MEM    TIME+  COMMAND
 729 root        20   0 435456 149672 58940 S   5.0   7.4   0:42.84 Xorg
1096 kali        20   0 458580 44904 34892 S   3.0   2.2   0:04.51 panel-1-whisker
1040 kali        20   0 975432 122228 82052 S   1.0   6.0   0:15.58 xfwm4
1100 kali        20   0 285908 47420 19200 S   0.7   2.3   0:12.53 panel-13-cpugra
 24 root        20   0 0 0 0 S   0.3   0.0   0:00.31 ksoftirqd/1
985 kali        20   0 215960 3212 2816 S   0.3   0.2   0:09.58 VBoxClient
1102 kali        20   0 337688 27560 20352 S   0.3   1.4   0:09.37 panel-15-genmon
10735 kali       20   0 458984 102200 88288 S   0.3   5.1   0:03.49 qterminal
  1 root        20   0 22684 14188 10204 S   0.0   0.7   0:00.68 systemd
  2 root        20   0 0 0 0 S   0.0   0.0   0:00.00 kthreadd
  3 root        20   0 0 0 0 S   0.0   0.0   0:00.00 pool_workqueue_release
  4 root        0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-rcu_g
  5 root        0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-rcu_p
  6 root        0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-slub_
  7 root        0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-netns
 11 root       20   0 0 0 0 I   0.0   0.0   0:00.00 kworker/u4:0-ext4-rsv-conversion
 12 root       0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-mm_pe
 13 root       20   0 0 0 0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
 14 root       20   0 0 0 0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
 15 root       20   0 0 0 0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
 16 root       20   0 0 0 0 S   0.0   0.0   0:00.41 ksoftirqd/0
 17 root       20   0 0 0 0 I   0.0   0.0   0:00.76 rcu_preempt
 18 root       rt   0 0 0 0 S   0.0   0.0   0:00.05 migration/0
 19 root      -51   0 0 0 0 S   0.0   0.0   0:00.00 idle_inject/0
 20 root       20   0 0 0 0 S   0.0   0.0   0:00.00 cpuhp/0
 21 root       20   0 0 0 0 S   0.0   0.0   0:00.00 cpuhp/1
 22 root      -51   0 0 0 0 S   0.0   0.0   0:00.00 idle_inject/1
 23 root       rt   0 0 0 0 S   0.0   0.0   0:00.18 migration/1
 26 root       0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/1:0H-events_highpri
 30 root       20   0 0 0 0 I   0.0   0.0   0:00.04 kworker/u6:1-flush-8:0
 31 root       20   0 0 0 0 S   0.0   0.0   0:00.00 kdevtmpfs
 32 root       0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-inet_
 33 root       20   0 0 0 0 S   0.0   0.0   0:00.00 kauditd
 35 root       20   0 0 0 0 S   0.0   0.0   0:00.00 khungtaskd
 36 root       20   0 0 0 0 S   0.0   0.0   0:00.00 oom_reaper
 38 root       0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-write
 39 root       20   0 0 0 0 S   0.0   0.0   0:00.19 kcompactd0
 40 root       25   5 0 0 0 S   0.0   0.0   0:00.00 ksmd
 41 root       39  19 0 0 0 S   0.0   0.0   0:00.43 khugepaged
 42 root       0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-kinte
 43 root       0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-kbloc
 44 root       0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-blkcg
 45 root      -51   0 0 0 0 S   0.0   0.0   0:00.00 irq/9-acpi
 46 root       0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-tpm_d
 47 root       0 -20 0 0 0 I   0.0   0.0   0:00.00 kworker/R-edac-

```



```
(kali㉿kali)-[~]
$ top -h

Usage:
top [options]

Options:
-b, --batch-mode           run in non-interactive batch mode
-c, --cmdline-toggle       reverse last remembered 'c' state
-d, --delay =SECS [.TENTHS] iterative delay as SECS [.TENTHS]
-E, --scale-summary-mem =SCALE set mem as: k,m,g,t,p,e for SCALE
-e, --scale-task-mem =SCALE  set mem with: k,m,g,t,p for SCALE
-H, --threads-show        show tasks plus all their threads
-i, --idle-toggle         reverse last remembered 'i' state
-n, --iterations =NUMBER  exit on maximum iterations NUMBER
-O, --list-fields          output all field names, then exit
-o, --sort-override =FIELD force sorting on this named FIELD
-p, --pid =PIDLIST        monitor only the tasks in PIDLIST
-S, --accum-time-toggle   reverse last remembered 'S' state
-s, --secure-mode         run with secure mode restrictions
-U, --filter-any-user =USER show only processes owned by USER
-u, --filter-only-euser =USER show only processes owned by USER
-w, --width [=COLUMNS]  change print width [,use COLUMNS]
-l, --single-cpu-toggle   reverse last remembered 'l' state

-h, --help                display this help text, then exit
-V, --version             output version information & exit

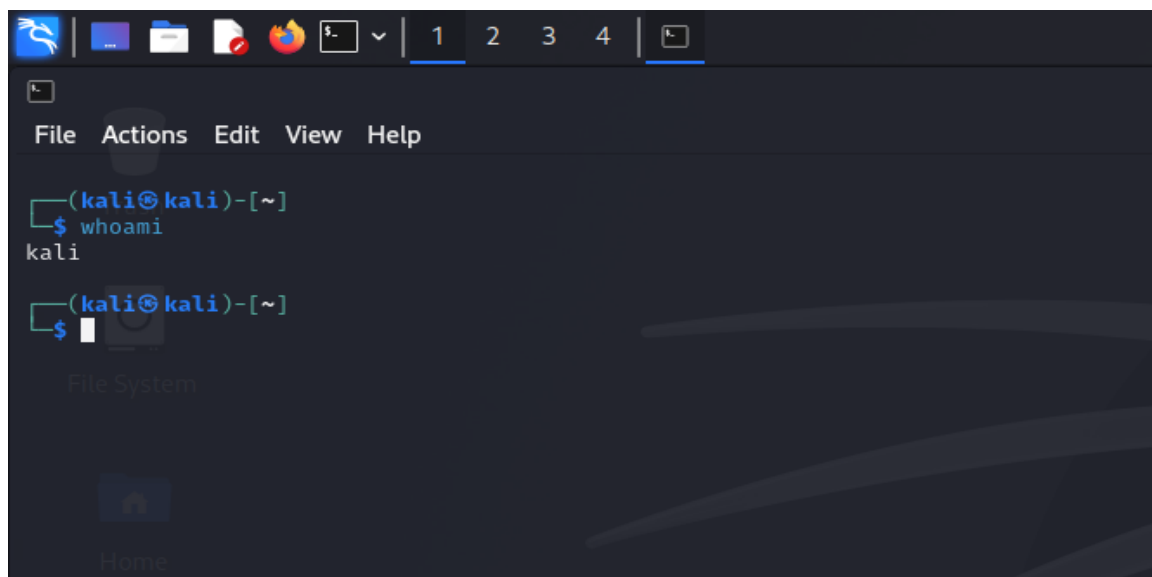
For more details see top(1).

(kali㉿kali)-[~]
$
```

15. whoami:

- The whoami command in Linux is used to display the **current user** who is logged into the system. It simply returns the username of the currently logged-in user. This command is useful when you want to confirm which user account you're operating under, especially when working with multiple user accounts or in scripts.

Implementation :



```
(kali㉿kali)-[~]
$ whoami
kali

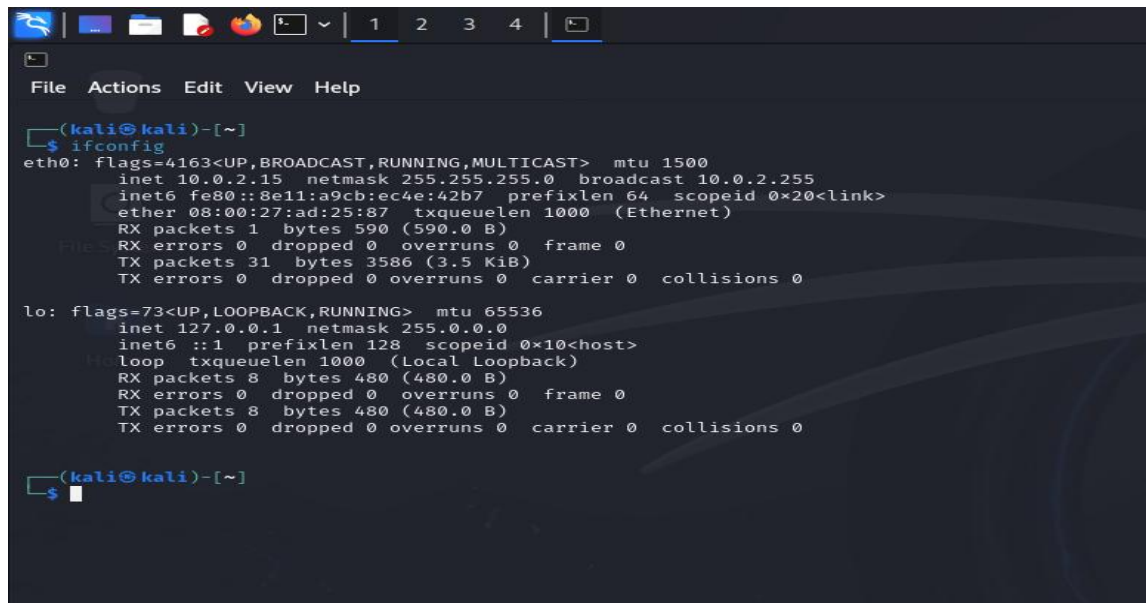
(kali㉿kali)-[~]
$
```

Date: 13/12/24

16. ifconfig:

- The ifconfig (interface configuration) command in Kali Linux is used to **configure, manage, and display network interfaces**. It shows the status of network interfaces, including their IP addresses, MAC addresses, and other related network configuration information.

Implementation :



```
(kali@kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::8e11:a9cb:ec4e:42b7 prefixlen 64 scopeid 0<link>
    ether 08:00:27:ad:25:87 txqueuelen 1000 (Ethernet)
    RX packets 1 bytes 590 (590.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 31 bytes 3586 (3.5 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

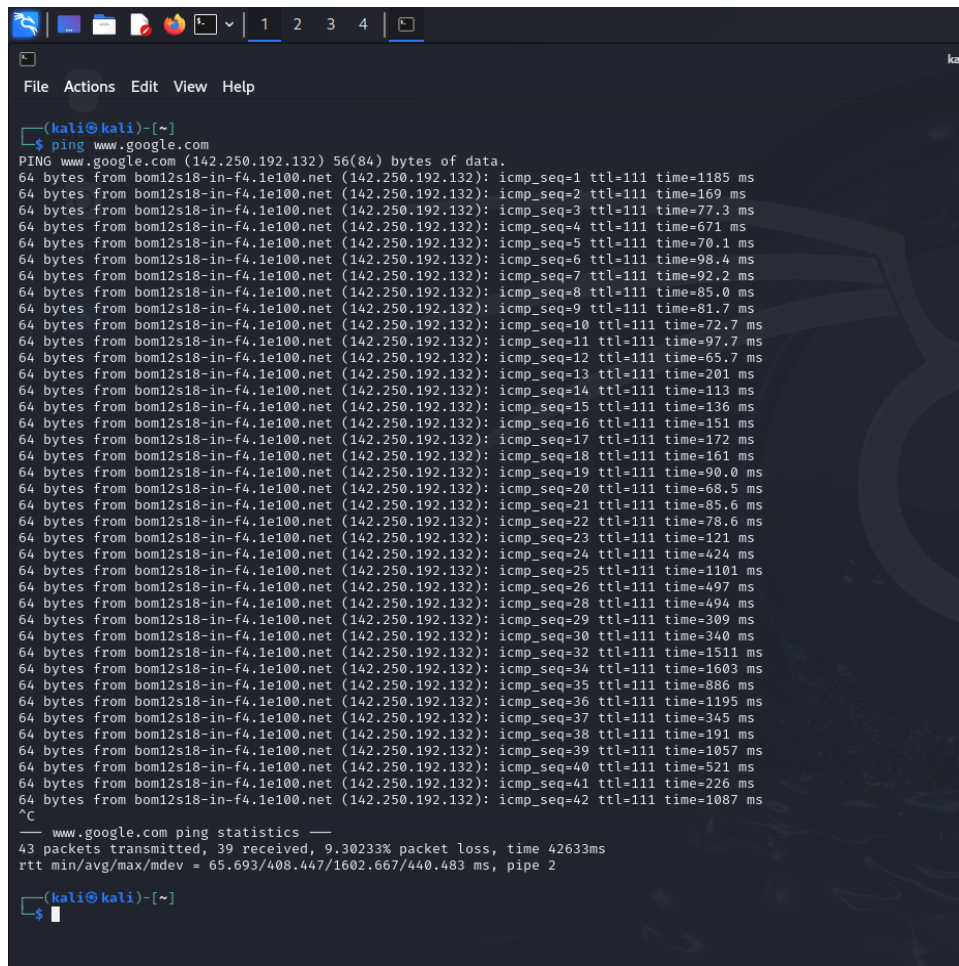
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(kali@kali)-[~]
$
```

17. ping:

- The ping command is a network utility used to test the connectivity between the local system and a remote host (IP address or domain name). It sends ICMP (Internet Control Message Protocol) echo request packets to the target and waits for an echo reply. It helps to determine if a network connection is working and how long it takes for packets to travel to the destination and back.

Implementation :



```
(kali@kali)-[~]
$ ping www.google.com
PING www.google.com (142.250.192.132) 56(84) bytes of data:
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=1 ttl=111 time=1185 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=2 ttl=111 time=169 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=3 ttl=111 time=77.3 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=4 ttl=111 time=671 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=5 ttl=111 time=70.1 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=6 ttl=111 time=98.4 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=7 ttl=111 time=92.2 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=8 ttl=111 time=85.0 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=9 ttl=111 time=81.7 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=10 ttl=111 time=72.7 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=11 ttl=111 time=97.7 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=12 ttl=111 time=65.7 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=13 ttl=111 time=201 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=14 ttl=111 time=113 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=15 ttl=111 time=136 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=16 ttl=111 time=151 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=17 ttl=111 time=172 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=18 ttl=111 time=161 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=19 ttl=111 time=90.0 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=20 ttl=111 time=68.5 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=21 ttl=111 time=85.6 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=22 ttl=111 time=78.6 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=23 ttl=111 time=121 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=24 ttl=111 time=424 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=25 ttl=111 time=1101 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=26 ttl=111 time=497 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=28 ttl=111 time=494 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=29 ttl=111 time=309 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=30 ttl=111 time=340 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=32 ttl=111 time=1511 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=34 ttl=111 time=1603 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=35 ttl=111 time=886 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=36 ttl=111 time=1195 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=37 ttl=111 time=345 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=38 ttl=111 time=191 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=39 ttl=111 time=1057 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=40 ttl=111 time=521 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=41 ttl=111 time=226 ms
64 bytes from bom12s18-in-f4.1e100.net (142.250.192.132): icmp_seq=42 ttl=111 time=1087 ms
^C
--- www.google.com ping statistics ---
43 packets transmitted, 39 received, 9.30233% packet loss, time 42633ms
rtt min/avg/max/mdev = 65.693/408.447/1602.667/440.483 ms, pipe 2
(kali@kali)-[~]
$
```

18. netstat:

- The netstat (network statistics) command in Kali Linux is used to display detailed information about **network connections**, **routing tables**, **interface statistics**, **masquerade connections**, and **multicast memberships**. It is a powerful tool for monitoring and troubleshooting network-related issues.

No.	Option	Description
1	netstat -a	Show all connections (listening and established).
2	netstat -t	Show TCP connections only.
3	netstat -u	Show UDP connections only.
4	netstat -l	Show only listening sockets.

Implementation :

```

(kali@kali)~$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
udp        0      0 10.0.2.15:bootpc       10.0.2.2:bootps        ESTABLISHED

Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags   Type       State         I-Node  Path
unix  3      [ ]     STREAM    CONNECTED    8613
unix  3      [ ]     STREAM    CONNECTED    8476
unix  3      [ ]     STREAM    CONNECTED    9449    @/tmp/.X11-unix/X0
unix  3      [ ]     STREAM    CONNECTED    7289    /run/systemd/journal/stdout
unix  3      [ ]     STREAM    CONNECTED    8603
unix  3      [ ]     STREAM    CONNECTED    8482
unix  3      [ ]     STREAM    CONNECTED    8458
unix  2      [ ]     DGRAM     CONNECTED    8892
unix  3      [ ]     STREAM    CONNECTED    8009    /run/user/1000/bus
unix  3      [ ]     STREAM    CONNECTED    7350    @/tmp/.X11-unix/X0
unix  3      [ ]     STREAM    CONNECTED    9622    @/tmp/.X11-unix/X0
unix  3      [ ]     STREAM    CONNECTED    8481    /run/user/1000/at-spi-bus_0
unix  3      [ ]     STREAM    CONNECTED    8446    @/tmp/.ICE-unix/908
unix  3      [ ]     STREAM    CONNECTED    8901
unix  3      [ ]     STREAM    CONNECTED    7164    /run/user/1000/bus
unix  3      [ ]     STREAM    CONNECTED    8611
unix  3      [ ]     STREAM    CONNECTED    8480
unix  3      [ ]     STREAM    CONNECTED    8461
unix  3      [ ]     STREAM    CONNECTED    8012
unix  3      [ ]     STREAM    CONNECTED    9654    /run/user/1000/bus
unix  3      [ ]     STREAM    CONNECTED    9471    @/tmp/.X11-unix/X0
unix  3      [ ]     STREAM    CONNECTED    9442    @/tmp/.X11-unix/X0
unix  3      [ ]     STREAM    CONNECTED    7349
unix  3      [ ]     STREAM    CONNECTED    8008
unix  3      [ ]     STREAM    CONNECTED    8905    /run/dbus/system_bus_socket
unix  3      [ ]     STREAM    CONNECTED    8445
unix  3      [ ]     STREAM    CONNECTED    8438    @/tmp/.ICE-unix/908
unix  3      [ ]     STREAM    CONNECTED    8612    /run/user/1000/at-spi-bus_0
unix  3      [ ]     STREAM    CONNECTED    9489    /run/user/1000/bus
unix  3      [ ]     STREAM    CONNECTED    7128    /run/dbus/system_bus_socket
unix  3      [ ]     DGRAM     CONNECTED    3843    /run/systemd/notify
unix  3      [ ]     STREAM    CONNECTED    9655    /run/user/1000/bus
unix  3      [ ]     STREAM    CONNECTED    8475    /run/user/1000/at-spi-bus_0
unix  3      [ ]     STREAM    CONNECTED    8457    /run/user/1000/at-spi-bus_0
unix  3      [ ]     STREAM    CONNECTED    8872
unix  3      [ ]     STREAM    CONNECTED    7937
unix  3      [ ]     STREAM    CONNECTED    8621
unix  3      [ ]     STREAM    CONNECTED    8471
unix  3      [ ]     STREAM    CONNECTED    9447    @/tmp/.X11-unix/X0
unix  3      [ ]     STREAM    CONNECTED    130619
unix  3      [ ]     STREAM    CONNECTED    7938    /run/systemd/journal/stdout
unix  3      [ ]     STREAM    CONNECTED    8624
unix  3      [ ]     STREAM    CONNECTED    9507    /run/user/1000/bus
  
```



```

(kali@kali)-[~]
$ netstat -a
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
udp        0      0 10.0.2.15:bootpc       10.0.2.2:bootps        ESTABLISHED
raw6       0      0 [::]:ipv6-icmp         [::]:*                  7

Active UNIX domain sockets (servers and established)
Proto RefCnt Flags       Type       State         I-Node   Path
unix   3      [ ]         STREAM     CONNECTED    8613     @/tmp/.X11-unix/X0
unix   3      [ ]         STREAM     CONNECTED    8476     @/tmp/.X11-unix/X0
unix   3      [ ]         STREAM     CONNECTED    9449     /run/systemd/journal/stdout
unix   3      [ ]         STREAM     CONNECTED    7289     /run/systemd/journal/stdout
unix   3      [ ]         STREAM     CONNECTED    7956     /run/systemd/journal/stdout
unix   3      [ ]         STREAM     CONNECTED    8603     /run/systemd/journal/stdout
unix   3      [ ]         STREAM     CONNECTED    8482     /run/systemd/journal/stdout
unix   3      [ ]         STREAM     CONNECTED    8458     /run/systemd/journal/stdout
unix   2      [ ]         DGRAM      CONNECTED    8892     /run/user/1000/bus
unix   3      [ ]         STREAM     CONNECTED    8009     /run/user/1000/bus
unix   3      [ ]         STREAM     CONNECTED    7350     @/tmp/.X11-unix/X0
unix   3      [ ]         STREAM     CONNECTED    9622     @/tmp/.X11-unix/X0
unix   3      [ ]         STREAM     CONNECTED    8481     /run/user/1000/at-spi/bus_0
unix   3      [ ]         STREAM     CONNECTED    8446     @/tmp/.ICE-unix/908
unix   3      [ ]         STREAM     CONNECTED    8901     /run/user/1000/bus
unix   3      [ ]         STREAM     CONNECTED    7164     /run/user/1000/bus
unix   3      [ ]         STREAM     CONNECTED    8611     /run/user/1000/bus
unix   3      [ ]         STREAM     CONNECTED    8480     /run/user/1000/bus
unix   3      [ ]         STREAM     CONNECTED    8461     /run/user/1000/bus
unix   3      [ ]         STREAM     CONNECTED    8012     /run/user/1000/bus
unix   3      [ ]         STREAM     CONNECTED    9654     /run/user/1000/bus
unix   3      [ ]         STREAM     CONNECTED    9471     @/tmp/.X11-unix/X0
unix   3      [ ]         STREAM     CONNECTED    9442     @/tmp/.X11-unix/X0
unix   3      [ ]         STREAM     CONNECTED    7349     /run/user/1000/bus
unix   3      [ ]         STREAM     CONNECTED    8008     /run/user/1000/bus
unix   3      [ ]         STREAM     CONNECTED    8905     /run/dbus/system_bus_socket
unix   3      [ ]         STREAM     CONNECTED    8445     /run/dbus/system_bus_socket
unix   3      [ ]         STREAM     CONNECTED    8438     @/tmp/.ICE-unix/908
unix   3      [ ]         STREAM     CONNECTED    8612     /run/user/1000/at-spi/bus_0
unix   3      [ ]         STREAM     CONNECTED    9489     /run/user/1000/bus
unix   2      [ ACC ]         STREAM     LISTENING    7287     /tmp/.X11-unix/X0
unix   3      [ ]         STREAM     CONNECTED    7128     /run/dbus/system_bus_socket
unix   3      [ ]         DGRAM      CONNECTED    3843     /run/systemd/notify
unix   2      [ ACC ]         STREAM     LISTENING    3848     /run/systemd/private
unix   3      [ ]         STREAM     CONNECTED    9655     /run/user/1000/bus
unix   3      [ ]         STREAM     CONNECTED    8475     /run/user/1000/at-spi/bus_0
unix   3      [ ]         STREAM     CONNECTED    8457     /run/user/1000/at-spi/bus_0
unix   3      [ ]         STREAM     CONNECTED    8872     /run/user/1000/at-spi/bus_0
unix   3      [ ]         STREAM     CONNECTED    7937     /run/user/1000/at-spi/bus_0
unix   2      [ ACC ]         STREAM     LISTENING    3849     /run/systemd/userdb/io.systemd.DynamicUser
unix   2      [ ACC ]         STREAM     LISTENING    8272     /tmp/ssh-8L8qoGp5DD0c/agent.908
unix   2      [ ACC ]         STREAM     LISTENING    3850     /run/systemd/io.systemd.ManagedOOM
unix   3      [ ]         STREAM     CONNECTED    8621

```

```

(kali@kali)-[~]
$ netstat -t
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State

(kali@kali)-[~]
$

```



```
(kali@kali)-[~]
$ netstat -t
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
(kali@kali)-[~]
$ netstat -u
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
udp        0      0 10.0.2.15:bootpc       10.0.2.2:bootps        ESTABLISHED
(kali@kali)-[~]
$
```

```
(kali@kali)-[~]
$ netstat -l
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
raw6     0      0 [::]:ipv6-icmp         [::]:*                  7
Active UNIX domain sockets (only servers)
Proto RefCnt Flags     Type       State           I-Node   Path
unix    2      [ ACC ] STREAM    LISTENING     7287     /tmp/.X11-unix/X0
unix    2      [ ACC ] STREAM    LISTENING     3848     /run/systemd/private
unix    2      [ ACC ] STREAM    LISTENING     3849     /run/systemd/userdb/io.systemd.DynamicUser
unix    2      [ ACC ] STREAM    LISTENING     8272     /tmp/ssh-8l8qoGp5DD0c/agent.908
unix    2      [ ACC ] STREAM    LISTENING     3850     /run/systemd/io.systemd.ManagedOOM
unix    2      [ ACC ] STREAM    LISTENING     9305     /tmp/.ICE-unix/908
unix    2      [ ACC ] STREAM    LISTENING     4675     /run/systemd/io.systemd.sysext
unix    2      [ ACC ] STREAM    LISTENING     3863     /run/systemd/io.systemd.Credentials
unix    2      [ ACC ] STREAM    LISTENING     3866     /run/systemd/journal/stdout
unix    2      [ ACC ] STREAM    LISTENING     3868     /run/udev/control
unix    2      [ ACC ] STREAM    LISTENING     7869     /run/user/1000/systemd/private
unix    2      [ ACC ] STREAM    LISTENING     7881     /run/user/1000/bus
unix    2      [ ACC ] STREAM    LISTENING     7882     /run/user/1000/gnupg/S.dirmngr
unix    2      [ ACC ] STREAM    LISTENING     7883     /run/user/1000/gcr/ssh
unix    2      [ ACC ] STREAM    LISTENING     7884     /run/user/1000/keyring/control
unix    2      [ ACC ] STREAM    LISTENING     7885     /run/user/1000/gnupg/S.gpg-agent.browser
unix    2      [ ACC ] STREAM    LISTENING     7886     /run/user/1000/gnupg/S.gpg-agent.extra
unix    2      [ ACC ] STREAM    LISTENING     7887     /run/user/1000/gnupg/S.gpg-agent.ssh
unix    2      [ ACC ] STREAM    LISTENING     7888     /run/user/1000/gnupg/S.gpg-agent
unix    2      [ ACC ] STREAM    LISTENING     7889     /run/user/1000/pulse/native
unix    2      [ ACC ] STREAM    LISTENING     7890     /run/user/1000/pipewire-0
unix    2      [ ACC ] STREAM    LISTENING     7891     /run/user/1000/pipewire-0-manager
unix    2      [ ACC ] STREAM    LISTENING     5594     /run/dbus/system_bus_socket
unix    2      [ ACC ] STREAM    LISTENING     5596     /run/pescd/pescd.comm
unix    2      [ ACC ] STREAM    LISTENING     5598     /run/ssh-unix-local/socket
unix    2      [ ACC ] STREAM    LISTENING     8203     /run/user/1000/keyring/pkcs11
unix    2      [ ACC ] STREAM    LISTENING     5600     /run/systemd/io.systemd.Hostname
unix    2      [ ACC ] STREAM    LISTENING     9271     /run/user/1000/at-spi/bus_0
unix    2      [ ACC ] STREAM    LISTENING     127554   /run/systemd/journal/io.systemd.journal
unix    2      [ ACC ] STREAM    LISTENING     7286     @/tmp/.X11-unix/X0
unix    2      [ ACC ] STREAM    LISTENING     9304     @/tmp/.ICE-unix/908
(kali@kali)-[~]
$
```

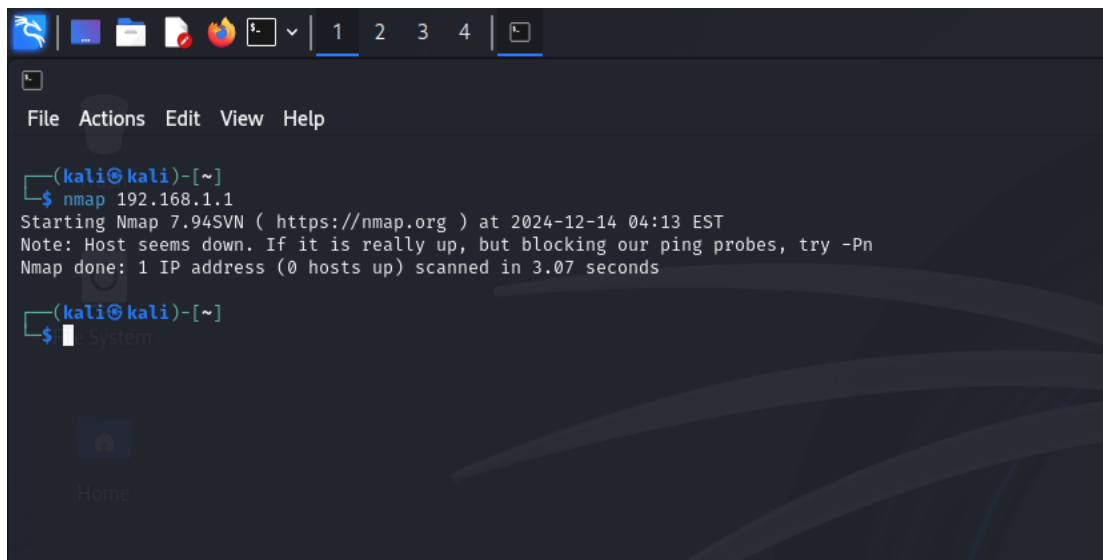
Date: 13/12/24

19. nmap:

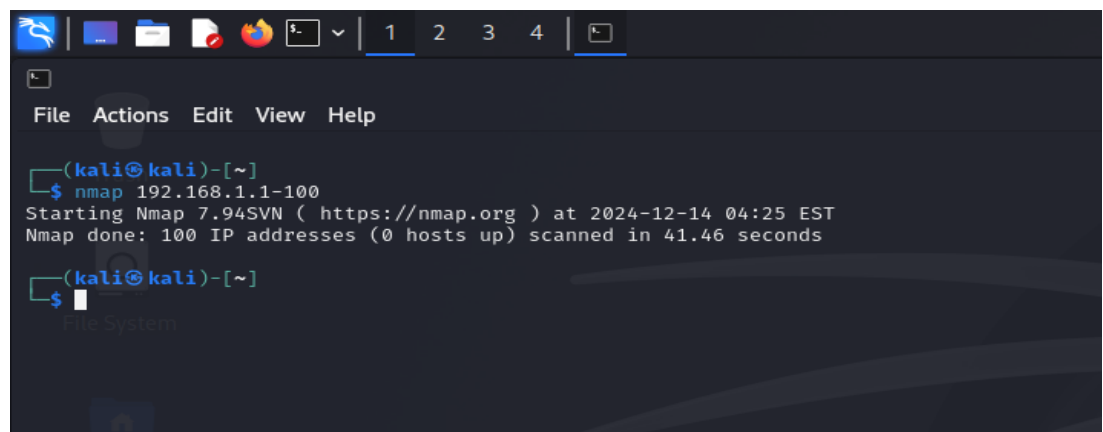
- The nmap (Network Mapper) command is a powerful network scanning tool used for discovering hosts and services on a computer network. It is often utilized by system administrators and security professionals for network inventory, security auditing, and vulnerability assessments.

No.	Option	Description
1	nmap 192.168.1.1	Scan a Single Host.
2	nmap 192.168.1.1-100	Scan a Range of Ips.
3	nmap 192.168.1.0/24	Scan a Subnet.

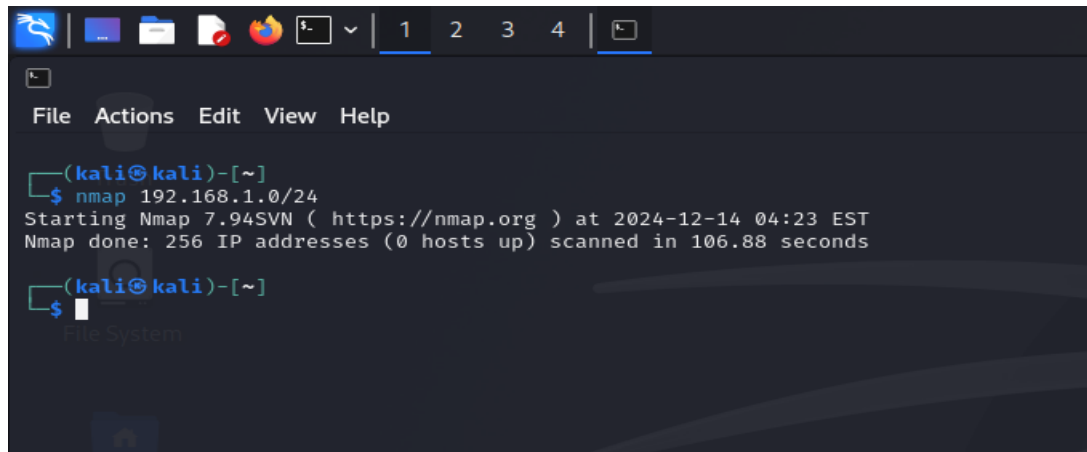
Implementation :



```
(kali㉿kali)-[~]
$ nmap 192.168.1.1
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-14 04:13 EST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.07 seconds
```



```
(kali㉿kali)-[~]
$ nmap 192.168.1.1-100
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-14 04:25 EST
Nmap done: 100 IP addresses (0 hosts up) scanned in 41.46 seconds
```



```
(kali@kali)-[~]
$ nmap 192.168.1.0/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-14 04:23 EST
Nmap done: 256 IP addresses (0 hosts up) scanned in 106.88 seconds

(kali@kali)-[~]
$
```

20. Wget:

- The wget command is a non-interactive network downloader in Linux. It allows users to download files from the internet using HTTP, HTTPS, and FTP protocols. Since it is non-interactive, it works seamlessly in scripts and runs in the background, making it ideal for automated tasks.

Implementation :



```
(kali@kali)-[~]
$ wget https://example.com/file.zip
--2024-12-14 04:31:26-- https://example.com/file.zip
Resolving example.com (example.com)... 93.184.215.14, 2606:2800:21f:cb07:6820:80da:af6b:8b2c
Connecting to example.com (example.com)[93.184.215.14]:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2024-12-14 04:31:28 ERROR 404: Not Found.

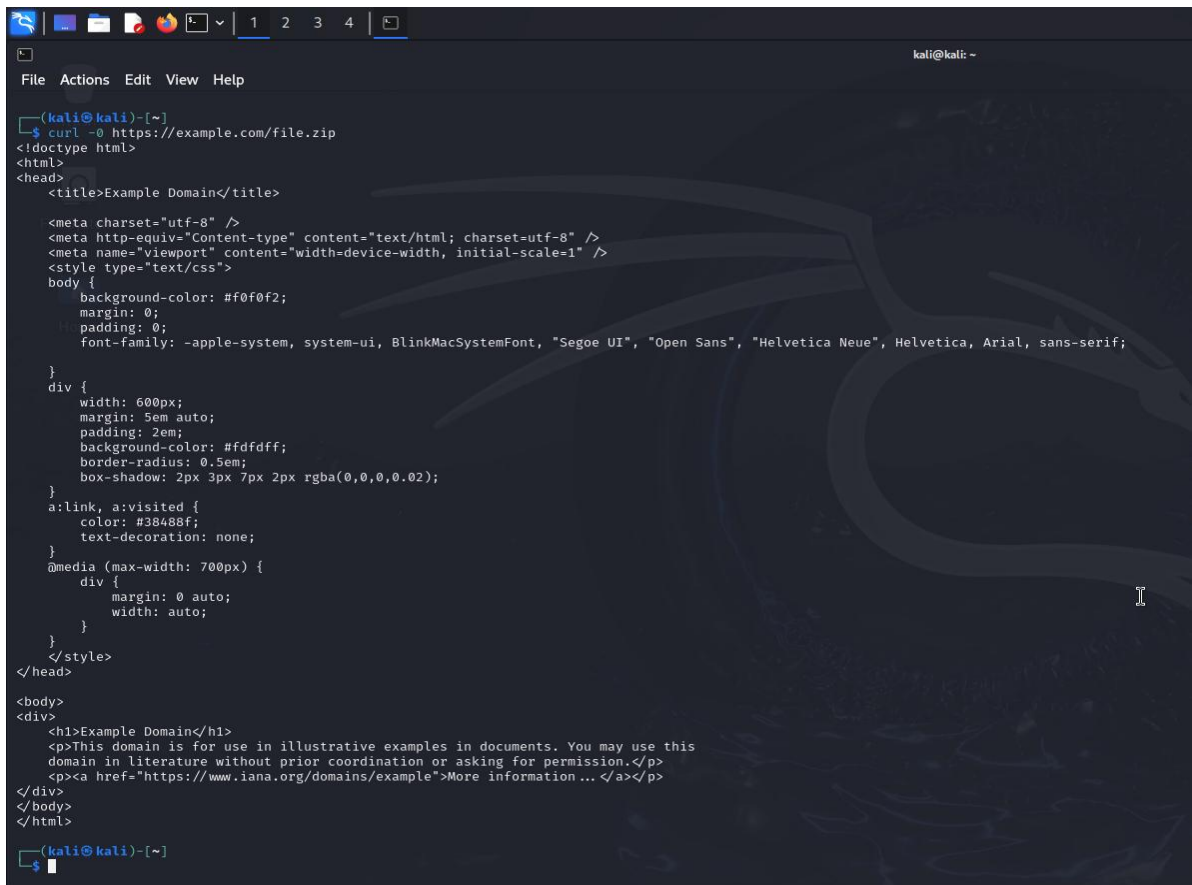
(kali@kali)-[~]
$
```

Date: 13/12/24

21. curl:

- The curl command is a powerful command-line tool for transferring data to or from a server. It supports a wide range of protocols, including HTTP, HTTPS, FTP, SCP, and more. curl is widely used for making API requests, testing endpoints, downloading files, and debugging network issues.

Implementation :



```
(kali@kali)~$ curl -o- https://example.com/file.zip
<!doctype html>
<html>
<head>
  <title>Example Domain</title>

  <meta charset="utf-8" />
  <meta http-equiv="Content-type" content="text/html; charset=utf-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1" />
  <style type="text/css">
    body {
      background-color: #f0f0f2;
      margin: 0;
      padding: 0;
      font-family: -apple-system, system-ui, BlinkMacSystemFont, "Segoe UI", "Open Sans", "Helvetica Neue", Helvetica, Arial, sans-serif;
    }
    div {
      width: 600px;
      margin: 5em auto;
      padding: 2em;
      background-color: #fdfdff;
      border-radius: 0.5em;
      box-shadow: 2px 3px 7px 2px rgba(0,0,0,0.02);
    }
    a:link, a:visited {
      color: #38488f;
      text-decoration: none;
    }
    @media (max-width: 700px) {
      div {
        margin: 0 auto;
        width: auto;
      }
    }
  </style>
</head>
<body>
<div>
  <h1>Example Domain</h1>
  <p>This domain is for use in illustrative examples in documents. You may use this
  domain in literature without prior coordination or asking for permission.</p>
  <p><a href="https://www.iana.org/domains/example">More information ...</a></p>
</div>
</body>
</html>

(kali@kali)~$
```

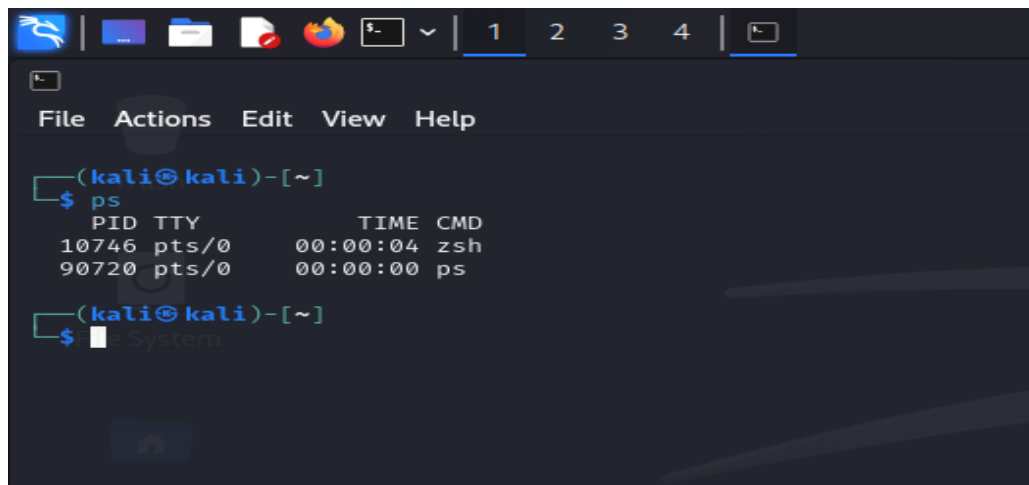
Date: 13/12/24

22. ps:

- The ps (process status) command in Linux is used to display information about currently running processes. It provides details such as process IDs, user ownership, CPU usage, and more.

No.	Option	Description
1	ps -e	Display All Processes.
2	ps -f	Display Processes in a Full Format

Implementation :



```
(kali@kali)-[~]
$ ps
  PID TTY          TIME CMD
 10746 pts/0    00:00:04 zsh
  90720 pts/0    00:00:00 ps
(kali@kali)-[~]
$
```

```
(kali㉿kali)-[~]
$ ps -e

```

PID	TTY	TIME	CMD
1 ?		00:00:00	systemd
2 ?		00:00:00	kthreadd
3 ?		00:00:00	pool_workqueue_release
4 ?		00:00:00	kworker/R-rcu_g
5 ?		00:00:00	kworker/R-rcu_p
6 ?		00:00:00	kworker/R-slub_
7 ?		00:00:00	kworker/R-netns
11 ?		00:00:00	kworker/u4:0-ext4-rsv-conversion
12 ?		00:00:00	kworker/R-mm_pe
13 ?		00:00:00	rcu_tasks_kthread
14 ?		00:00:00	rcu_tasks_rude_kthread
15 ?		00:00:00	rcu_tasks_trace_kthread
16 ?		00:00:00	ksoftirqd/0
17 ?		00:00:01	rcu_preempt
18 ?		00:00:00	migration/0
19 ?		00:00:00	idle_inject/0
20 ?		00:00:00	cpuhp/0
21 ?		00:00:00	cpuhp/1
22 ?		00:00:00	idle_inject/1
23 ?		00:00:00	migration/1
24 ?		00:00:00	ksoftirqd/1
26 ?		00:00:00	kworker/1:0H-events_highpri
31 ?		00:00:00	kdevtmpfs
32 ?		00:00:00	kworker/R-inet_
33 ?		00:00:00	kauditd
35 ?		00:00:00	khungtaskd
36 ?		00:00:00	oom_reaper
38 ?		00:00:00	kworker/R-write
39 ?		00:00:00	kcompactd0
40 ?		00:00:00	ksmd
41 ?		00:00:00	khugepaged
42 ?		00:00:00	kworker/R-kinte
43 ?		00:00:00	kworker/R-kbloc
44 ?		00:00:00	kworker/R-blkcg
45 ?		00:00:00	irq/9-acpi
46 ?		00:00:00	kworker/R-tpm_d
47 ?		00:00:00	kworker/R-edac-
48 ?		00:00:00	kworker/R-devfr
49 ?		00:00:00	kworker/0:1H-kblockd
50 ?		00:00:00	kswapd0
57 ?		00:00:00	kworker/R-kthro
59 ?		00:00:00	kworker/R-acpi_
60 ?		00:00:00	kworker/R-mld
61 ?		00:00:00	kworker/R-ipv6_
66 ?		00:00:00	kworker/R-kstrp
70 ?		00:00:00	kworker/u7:0
71 ?		00:00:00	kworker/u8:0-ttm
72 ?		00:00:00	kworker/u9:0

```
(kali㉿kali)-[~]
$ ps -f

```

UID	PID	PPID	C	STIME	TTY	TIME	CMD
kali	10746	10735	0	00:34	pts/0	00:00:04	/usr/bin/zsh
kali	95052	10746	0	04:58	pts/0	00:00:00	ps -f

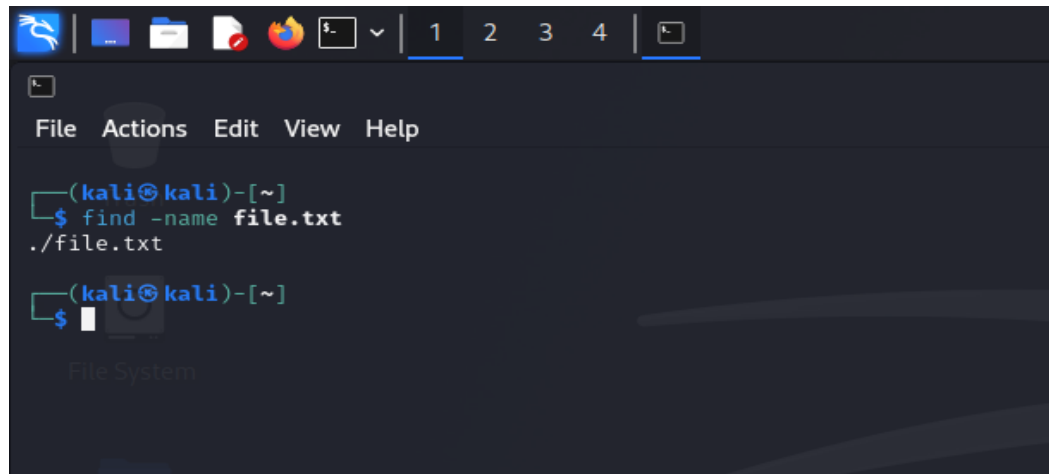
```
(kali㉿kali)-[~]
$ System
```

Date: 13/12/24

23. find:

- The find command in Linux is used to search for files and directories within a specified location. It can search based on a variety of criteria like name, type, size, modification time, and more.

Implementation :



```
(kali㉿kali)-[~]  
$ find -name file.txt  
./file.txt  
  
(kali㉿kali)-[~]  
$
```